北京交通大學

信号与系统

School of Electronic
and Information Engineering

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2025.3

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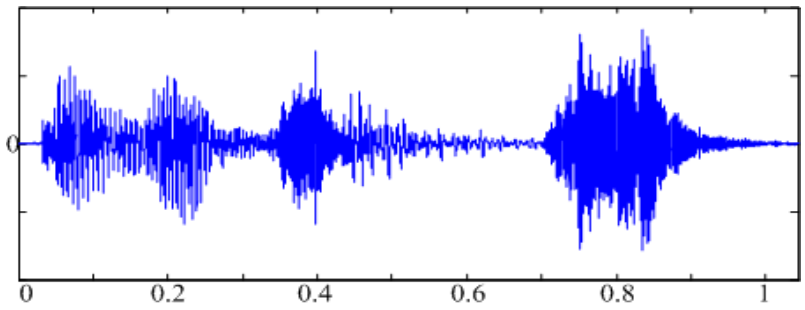
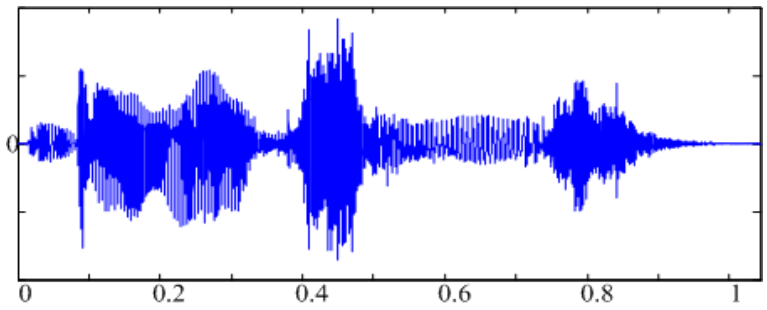
Why do we introduce frequency domain analysis of signals?

How to analyze the characteristics of male and female voices?



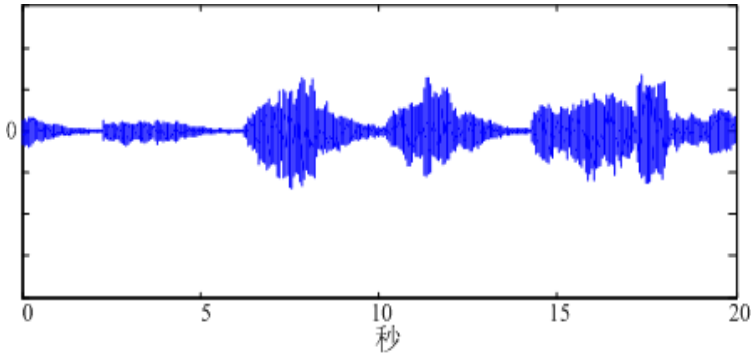
Time domain waveform of female voice



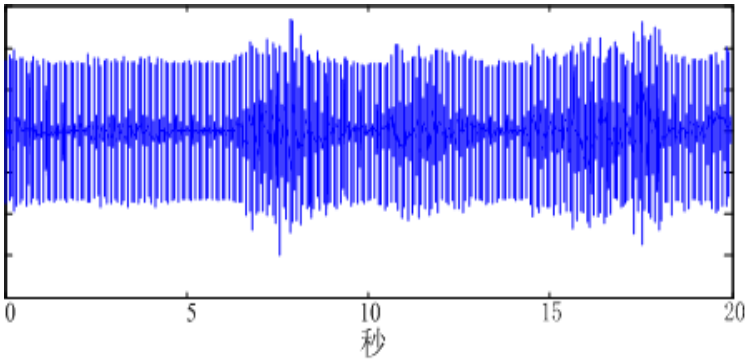
Time domain waveform of male voice

Why do we introduce frequency domain analysis of signals?

How to filter out the noise in the signal?

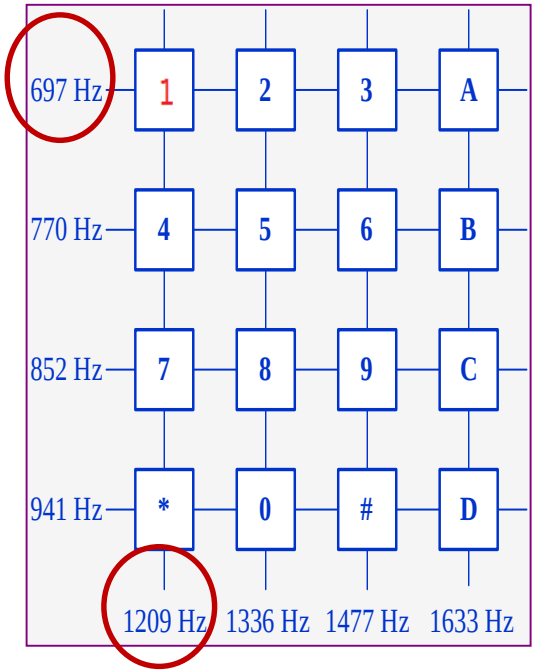


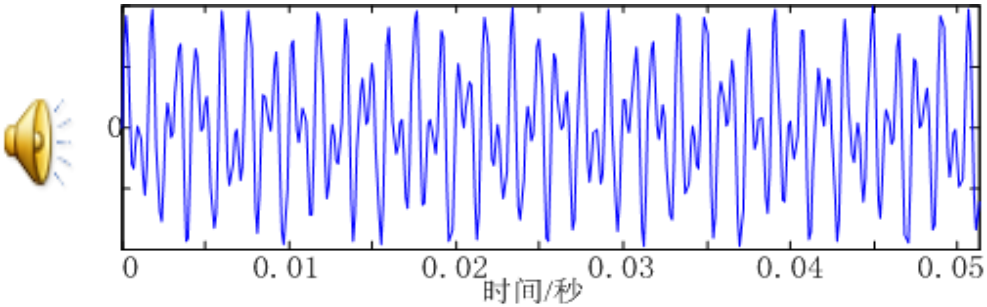
The time domain waveform of the signal



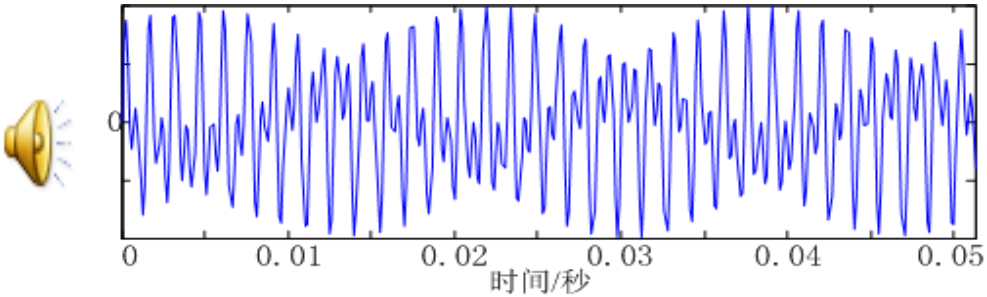
Time domain waveform of a noisy signal

Why do we introduce frequency domain analysis of signals?





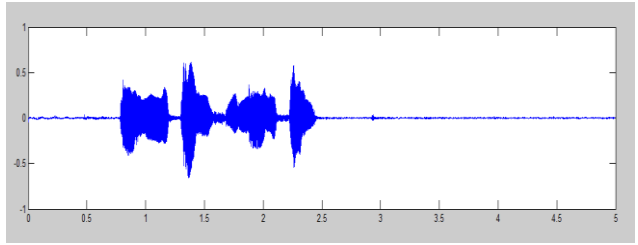
Time domain waveform of the number key '1'

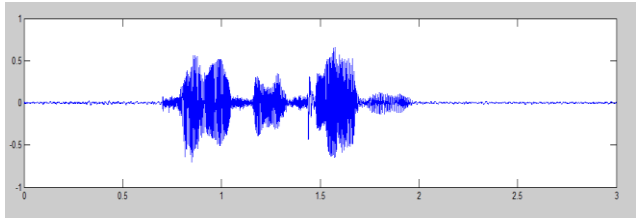


Time domain waveform of the number key ‘2'

Why do we introduce frequency domain analysis of signals?

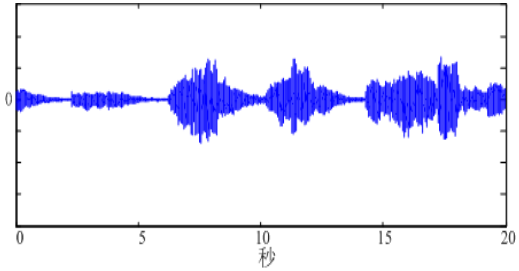
Analyze the voices of girls and boys

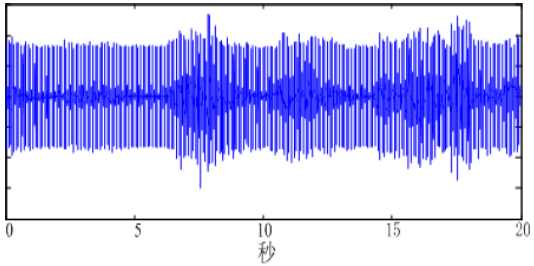




Time domain waveform of girl/boy voice

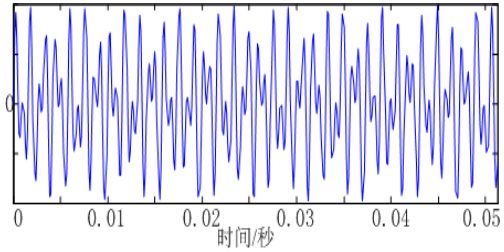
Filter out the noise in the signal

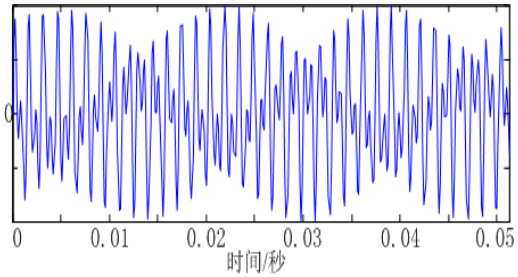




Signal/noisy signal time domain waveform

Recognize number keys





Digital keys' 1 'and' 2 'time domain waveforms

There are limitations in signal time domain analysis and processing

Frequency domain analysis of signals

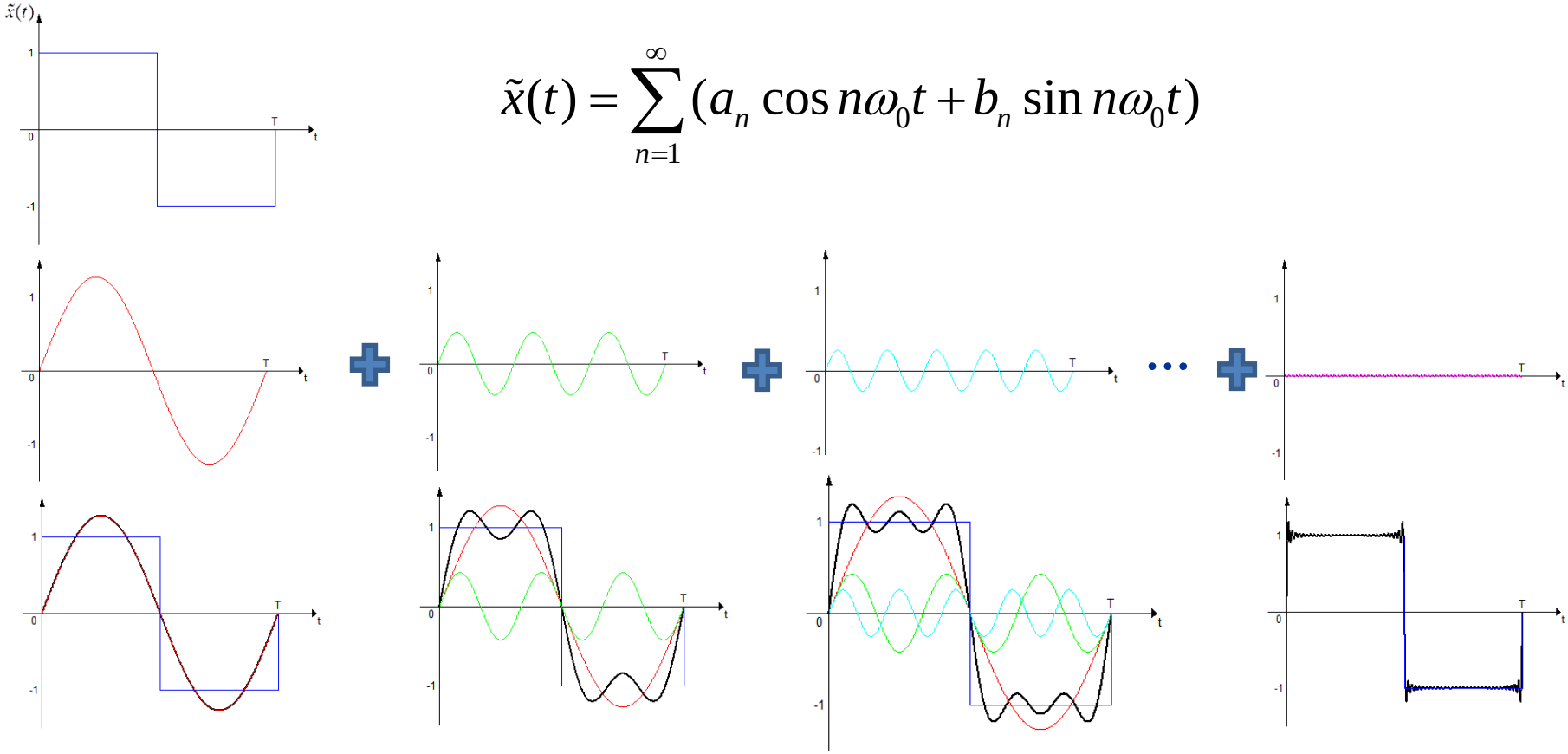
- Frequency Domain analysis of continuous periodic signals
- Frequency Domain analysis of continuous aperiodic signals
- Frequency Domain analysis of discrete periodic signals
- Frequency Domain analysis of discrete aperiodic signals
- Time-domain sampling of continuous-time signals
- Analyze spectrums of signals via MATLAB

Frequency Domain analysis of continuous periodic signals

- ※ Frequency domain representation of periodic signals
- ※ Spectrum of a periodic signal
- ※ Basic properties of Fourier series
- ※ Power spectrum of a periodic signal

Representation of periodic signals in frequency domain

$$\tilde{x}(t) = \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)$$



Fourier, Jean Baptiste Joseph (1768-1830)

Fourier was a French mathematician and physicist, whose main contribution was the development of a mathematical theory while studying the propagation of heat. In 1807, he submitted a paper titled "The Propagation of Heat" to the Paris Academy of Sciences, where he derived the famous heat conduction equation. While solving this equation, he discovered that the solution function could be represented by a series of trigonometric functions, thus proposing that any function could be expanded into an infinite series of trigonometric functions. In his seminal work "The Analytical Theory of Heat" published in 1822, he addressed the problem of heat distribution and propagation in non-uniformly heated solids, marking one of the earliest applications of analysis in physics. This had a profound impact on the development of mathematics and theoretical physics in the 19th century. The theories of Fourier series (i.e., trigonometric series) and Fourier analysis originated from this work.

The Proposal of Fourier Series

Fourier

×

Lagrange

√

Laplace

Lacroix

Monge

Representation of periodic signals in frequency domain

※ Fourier series in exponential form

$$\tilde{x}(t) = \sum_{n=-\infty}^{\infty} C_n \, \mathrm{e}^{\mathrm{j}n\omega_0 t}$$

$$\tilde{x}(t) = C_0 + C_{-1}\mathrm{e}^{-\mathrm{j}\omega_0 t} + C_1\mathrm{e}^{\mathrm{j}\omega_0 t} + C_{-2}\mathrm{e}^{-\mathrm{j}2\omega_0 t} + C_2\mathrm{e}^{\mathrm{j}2\omega_0 t} + \ldots + C_{-N}\mathrm{e}^{-\mathrm{j}N\omega_0 t} + C_N\mathrm{e}^{\mathrm{j}N\omega_0 t} + \ldots$$

DC component

Fundamental wave component

N harmonic component

- A periodic signal $\tilde{x}(t)$ can be represented as a **linear superposition** of infinite imaginary exponential signals
- T_0 is the period of the periodic signal $\tilde{x}(t)$ and ω_0 is the angular frequency
- C_n is a function of $n\omega_0$, $C_n = C_n(n\omega_0)$ which is abbreviated as C_n

Representation of periodic signals in frequency domain

A periodic signal $\tilde{x}(t)$ is expanded as a Fourier series (Dirichlet) condition:

(1) It is absolutely integrable in one period, that is, it satisfies

$$\int_{-T_0/2}^{T_0/2} \left|\tilde{x}(t)\right| \mathrm{d}t < \infty$$

(2) there are only finite discontinuities in a period;

(3) There are only finitely many maxima and minima in a period.

Representation of periodic signals in frequency domain

$$\tilde{x}(t)=\sum_{n=-\infty}^{\infty} C_n \, \mathrm{e}^{\mathrm{j}n\omega_0 t}$$

$$\tilde{x}(t)=\ldots+C_{-N}\mathrm{e}^{-\mathrm{j}N\omega_0 t}+\ldots+C_{-1}\mathrm{e}^{-\mathrm{j}\omega_0 t}+C_0+C_1\mathrm{e}^{\mathrm{j}\omega_0 t}+\ldots+C_N\mathrm{e}^{\mathrm{j}N\omega_0 t}+\ldots$$

※The period of $\mathrm{e}^{\pm \mathrm{j}N\omega_0 t}$ is T_0/N, the set of functions is **orthogonal** to each other on the interval [0, T_0].

$$\int_0^{T_0} \mathrm{e}^{\mathrm{j}N\omega_0 t}\mathrm{e}^{-\mathrm{j}M\omega_0 t}\mathrm{d}t=\begin{cases} T_0, & N=M \\ 0, & N\neq M \end{cases}$$

$$C_n=\frac{1}{T_0}\int_0^{T_0}\tilde{x}(t)\mathrm{e}^{-\mathrm{j}n\omega_0 t}\mathrm{d}t=\frac{1}{T_0}\int_{t_1}^{t_1+T_0}\tilde{x}(t)\mathrm{e}^{-\mathrm{j}n\omega_0 t}\mathrm{d}t$$

Orthogonal decomposition and Fourier series

◆ A set of orthogonal functions $\{\cdots,\ f_1(t),\ f_2(t),\ f_3(t),\ \cdots\}$ is orthogonal to each other.

$$\int_{t_1}^{t_2} f_i(t)\, f_j^*(t)\mathrm{d}t = \begin{cases} m & i = j \\ 0 & i \neq j \end{cases}$$

◆ The signal $x(t)$ can be represented by the set of orthogonal functions as follows.

$$x(t) = \sum_{n=-\infty}^{\infty} c_n f_n(x) = \cdots + c_1 f_1(x) + c_2 f_2(x) + \ldots + c_N f_N(x) + \cdots, \qquad t_1 < t < t_2$$

$$c_n = \frac{\int_{t_1}^{t_2} x(t)\, f_n^*(t)\mathrm{d}t}{\int_{t_1}^{t_2} f_n(t)\, f_n^*(t)\mathrm{d}t}$$ (Basis function)

Orthogonal decomposition and Fourier series

- If only finite terms are used to represent the signal $x(t)$, its approximation can be expressed as follows:

$$x(t) \approx c_1 f_1(t) + c_2 f_2(t) + \ldots + c_N f_N(t) = \sum_{n=1}^{N} c_n f_n(t) = \hat{x}(t)$$

$$\mathrm{MSE} = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} [x(t) - \hat{x}(t)]^2 \mathrm{d}t = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} [x(t) - \sum_{n=1}^{N} c_n f_n(t)]^2 \mathrm{d}t$$

- The mean square error (MSE) between the signal $\hat{x}(t)$ approximated by the set of orthogonal functions and the signal $x(t)$ is minimized when the following coefficients are chosen

$$c_n = \frac{\int_{t_1}^{t_2} x(t) f_n^*(t) \mathrm{d}t}{\int_{t_1}^{t_2} f_n(t) f_n^*(t) \mathrm{d}t}$$

Orthogonal decomposition and Fourier series

$$x(t)=\sum_{n=-\infty}^{\infty} c_n f_n(x)=\cdots+c_1 f_1(x)+c_2 f_2(x)+\ldots+c_N f_N(x)+\cdots, \qquad t_1<t<t_2$$

When $f_n(t)=\mathrm{e}^{\mathrm{j}n\omega_0 t}$ $\quad \tilde{x}(t)=\sum_{n=-\infty}^{\infty} C_n\ \mathrm{e}^{\mathrm{j}n\omega_0 t}$

According to the minimum mean square error criterion, C_n can be obtained

$$C_n=\frac{\int_{t_1}^{t_2} x(t) f_n^*(t)\mathrm{d}t}{\int_{t_1}^{t_2} f_n(t) f_n^*(t)\mathrm{d}t}=\frac{\int_{t_1}^{t_1+T_0} x(t)\mathrm{e}^{-\mathrm{j}n\omega_0 t}\mathrm{d}t}{\int_{t_1}^{t_1+T_0} \mathrm{e}^{\mathrm{j}n\omega_0 t}\mathrm{e}^{-\mathrm{j}n\omega_0 t}\mathrm{d}t}=\frac{1}{T_0}\int_{t_1}^{t_1+T_0} x(t)\mathrm{e}^{-\mathrm{j}n\omega_0 t}\mathrm{d}t$$

※ Fourier series in triangular form

If $\tilde{x}(t)$ is real signal, then C_n has conjugate even symmetry.

i.e. $C_n = C_{-n}^*$

$$\tilde{x}(t) = \sum_{n=-\infty}^{\infty} C_n \, \mathrm{e}^{\mathrm{j}n\omega_0 t} = C_0 + \sum_{n=-\infty}^{-1} C_n \mathrm{e}^{\mathrm{j}n\omega_0 t} + \sum_{n=1}^{\infty} C_n \mathrm{e}^{\mathrm{j}n\omega_0 t}$$

$$= C_0 + \sum_{n=1}^{\infty} \left(C_n \mathrm{e}^{\mathrm{j}n\omega_0 t} + C_{-n} \mathrm{e}^{-\mathrm{j}n\omega_0 t} \right)$$

If $C_n = \dfrac{a_n - \mathrm{j}b_n}{2}$ Then $C_{-n} = \dfrac{a_n + \mathrm{j}b_n}{2}$

※ Fourier series in triangular form

$$\tilde{x}(t)=\sum_{n=-\infty}^{\infty} C_n \, \mathrm{e}^{\mathrm{j}n\omega_0 t}=\frac{a_0}{2}+\sum_{n=1}^{\infty}(a_n \cos n\omega_0 t+b_n \sin n\omega_0 t)$$

Among them: $a_0=\frac{2}{T_0}\int_{t_0}^{t_0+T_0}\tilde{x}(t)\mathrm{d}t$

$$a_n=\frac{2}{T_0}\int_{t_0}^{t_0+T_0}\tilde{x}(t)\cos(n\omega_0 t)\mathrm{d}t \qquad (n=1,2\cdots)$$

$$b_n=\frac{2}{T_0}\int_{t_0}^{t_0+T_0}\tilde{x}(t)\sin(n\omega_0 t)\mathrm{d}t \qquad (n=1,2\cdots)$$

※ Fourier series in triangular form

$$\tilde{x}(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty}(a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)$$

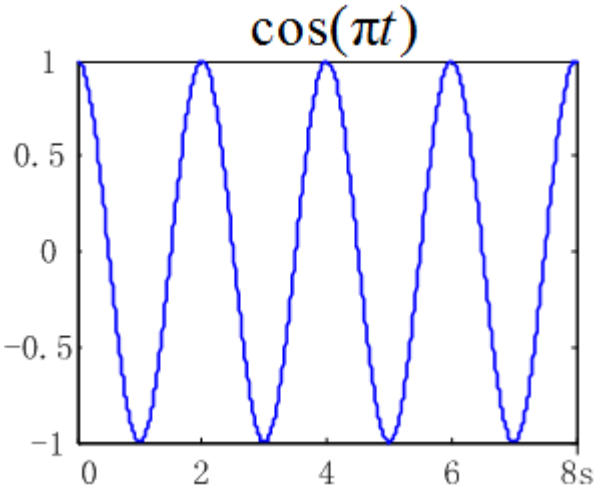
$$= \frac{a_0}{2} + \sum_{n=1}^{\infty} A_n \cos(n\omega_0 t + \phi_n)$$ (Pure cosine form)

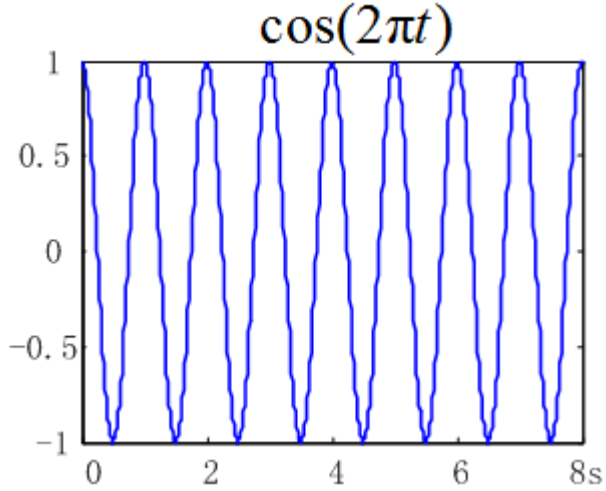
Where $A_n = \sqrt{a_n^2 + b_n^2}$ $\phi_n = \arctan\left(\frac{b_n}{a_n}\right)$

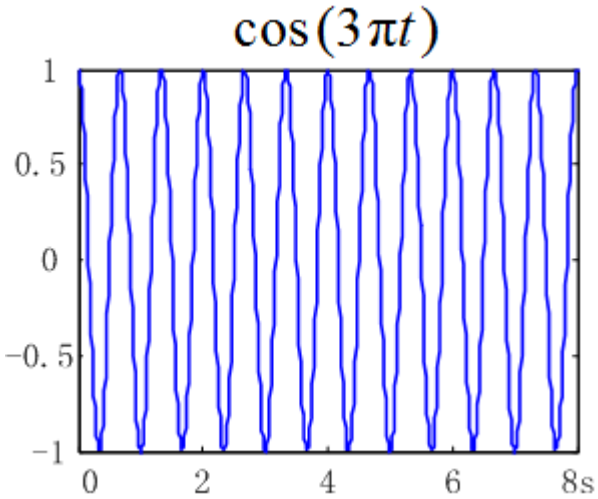
$a_0/2$ is called the DC component of the signal,

$A_n \cos(n\omega_0 t + f_n)$ is called the *N harmonic component* of the signal.

Example of Representing Other Signals with Sine Waves of Different Frequencies

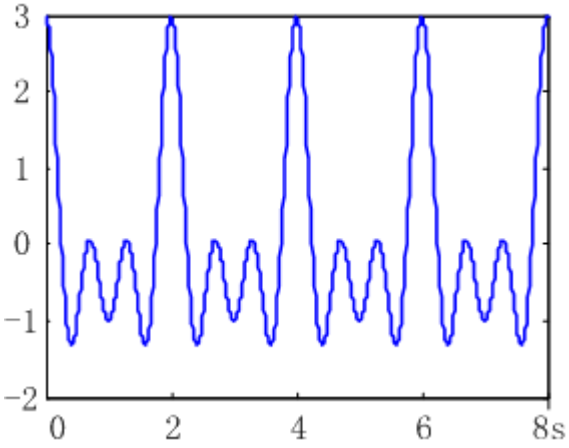






$x(t),\quad a_1 = a_2 = a_3 = 1$

$$x(t) = a_1\cos(\pi t) + a_2\cos(2\pi t) + a_3\cos(3\pi t)$$

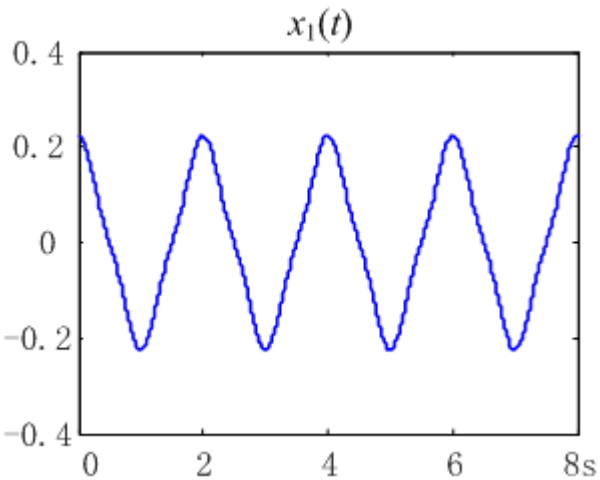


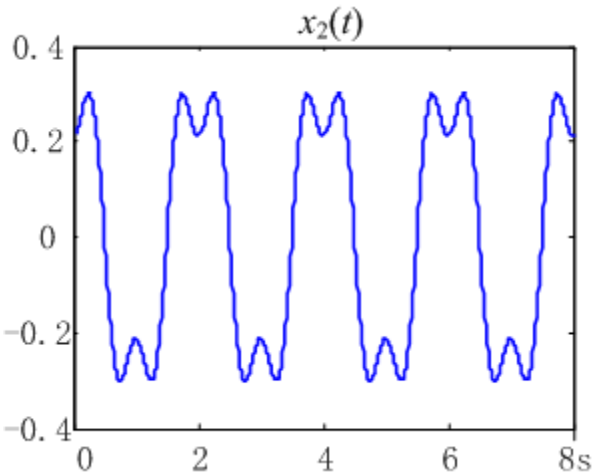
***Change the amplitudes a_1, a_2, a_3 of sine waves*, observe the changes in $x(t)$.**

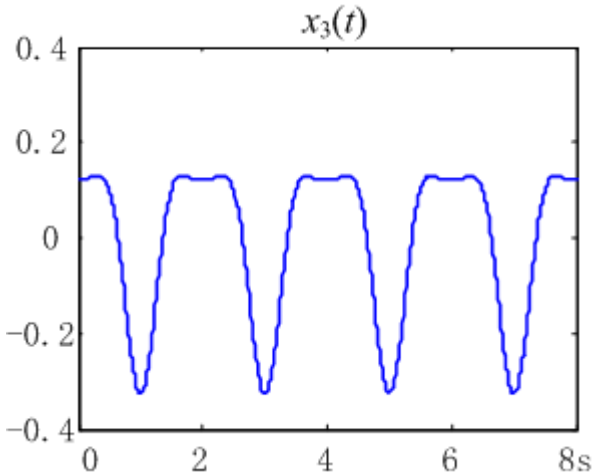
Example of Representing Other Signals with Sine Waves of Different Frequencies

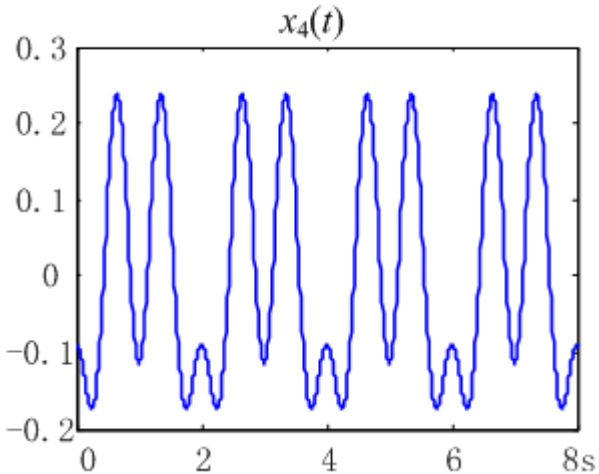
$$x(t) = a_1 \cos(\pi t) + a_2 \cos(2\pi t) + a_3 \cos(3\pi t)$$

$x(t)$	a_1	a_2	a_3
$x_1(t)$	0.3183	0.0000	–0.1061
$x_2(t)$	0.2026	0.0000	0.0225
$x_3(t)$	0.2026	–0.1013	0.0225
$x_4(t)$	–0.1157	–0.1013	0.1286









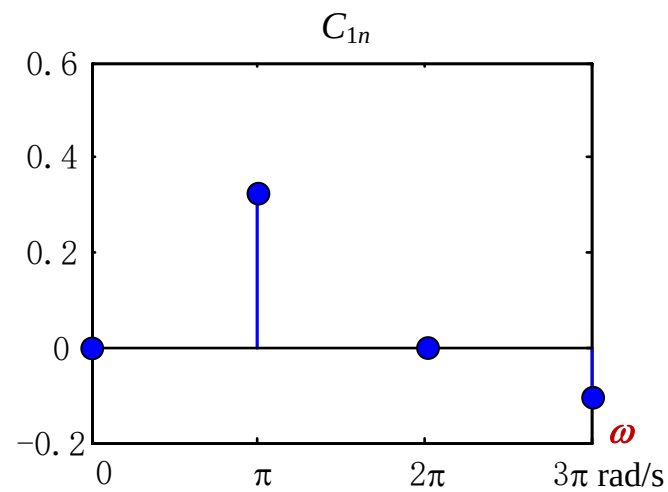
The waveform of $x(t)$ changes with different amplitudes of sine waves.

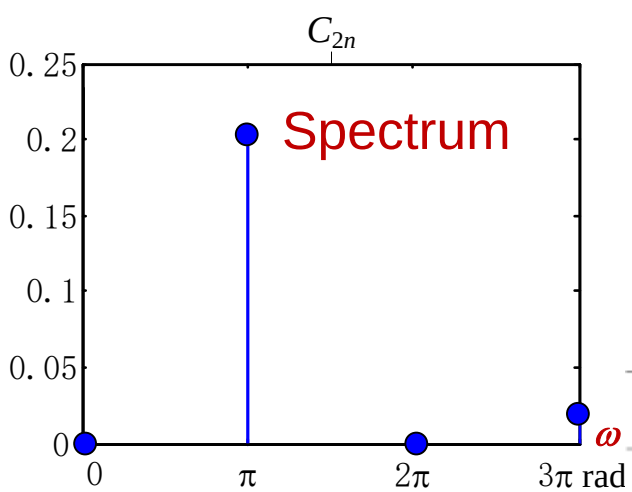
Example of Representing Other Signals with Sine Waves of Different Frequencies

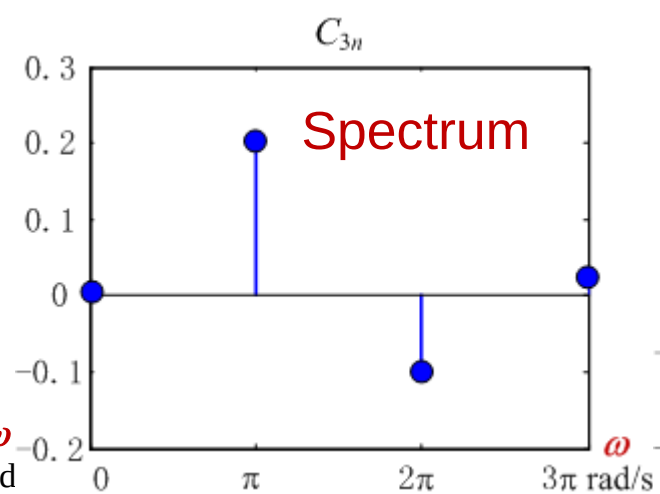
$$x(t) = a_1 \cos(\pi t) + a_2 \cos(2\pi t) + a_3 \cos(3\pi t)$$

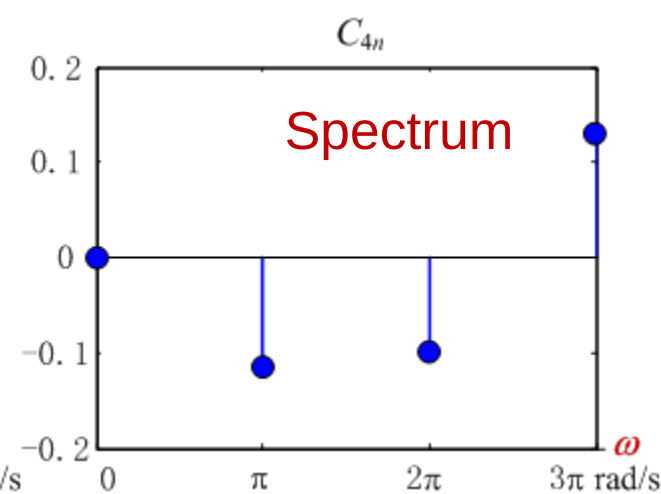
Coefficient \ ω	π	2π	3π
C_{1n}	0.3183	0.0000	–0.1061
C_{2n}	0.2026	0.0000	0.0225
C_{3n}	0.2026	–0.1013	0.0225
C_{4n}	–0.1157	–0.1013	0.1286

Plot the amplitudes of each sine wave with frequency as the variable.









Spectrum of Continuous Periodic Signals

$$\tilde{x}(t) = \sum_{n=-\infty}^{\infty} C_n \, \mathrm{e}^{\mathrm{j}n\omega_0 t}$$

Fourier series in exponential form

$$\Updownarrow$$

$$C_n = \frac{1}{T_0}\int_0^{T_0} \tilde{x}(t)\mathrm{e}^{-\mathrm{j}n\omega_0 t}\mathrm{d}t$$

- A periodic signal $\tilde{x}(t)$ can be represented as a **linear superposition** of infinite imaginary exponential signals

- Cn is a function of $n\omega_0$, $C_n = C_n(n\omega_0)$ which is abbreviated as Cn

Spectrum of Continuous Periodic Signals

$$\tilde{x}(t) = \sum_{n=-\infty}^{\infty} C_n \, \mathrm{e}^{\mathrm{j}n\omega_0 t}$$

$$\tilde{x}(t) = C_0 + C_{-1}\mathrm{e}^{-\mathrm{j}\omega_0 t} + C_1\mathrm{e}^{\mathrm{j}\omega_0 t} + C_{-2}\mathrm{e}^{-\mathrm{j}2\omega_0 t} + C_2\mathrm{e}^{\mathrm{j}2\omega_0 t} + \ldots + C_{-N}\mathrm{e}^{-\mathrm{j}N\omega_0 t} + C_N\mathrm{e}^{\mathrm{j}N\omega_0 t} + \ldots$$

DC component

Fundamental wave component

N harmonic component

- C_n reflects the distribution of each harmonic in the periodic signal $\tilde{x}(t)$

$\boldsymbol{C_n}$ is called the spectrum of the periodic signal $\tilde{x}(t)$, and there is a one-to-one correspondence with $\boldsymbol{C_n}$.

Spectrum of Continuous Periodic Signals

$$\tilde{x}(t) = \sum_{n=-\infty}^{\infty} C_n \, \mathrm{e}^{\mathrm{j}n\omega_0 t}$$

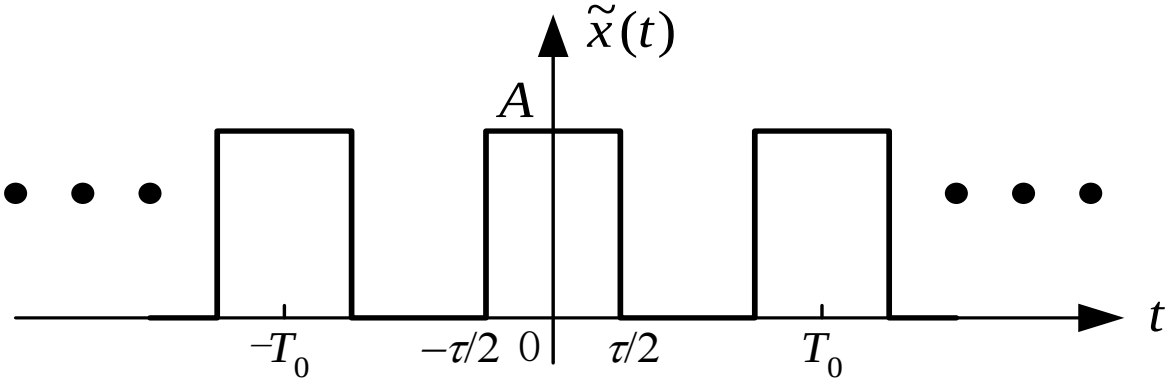
$$C_n = \frac{1}{T_0} \int_{t_0}^{T_0+t_0} \tilde{x}(t) \mathrm{e}^{-\mathrm{j}n\omega_0 t} \mathrm{d}t$$

$$C_n = \underline{|C_n|} \, \mathrm{e}^{\mathrm{j}\underline{\varphi_n}}$$

Phase spectrum

Amplitude spectrum

Ex. 1 Try to calculate the periodic rectangular pulse signal $\tilde{x}(t)$ shown in the figure.



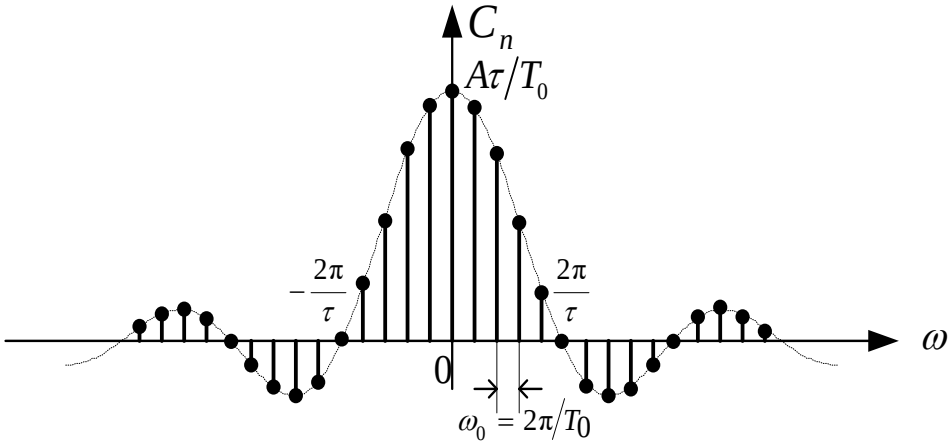
Solution:

$$C_n = \frac{1}{T_0}\int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} \tilde{x}(t)\mathrm{e}^{-\mathrm{j}n\omega_0 t}\mathrm{d}t$$

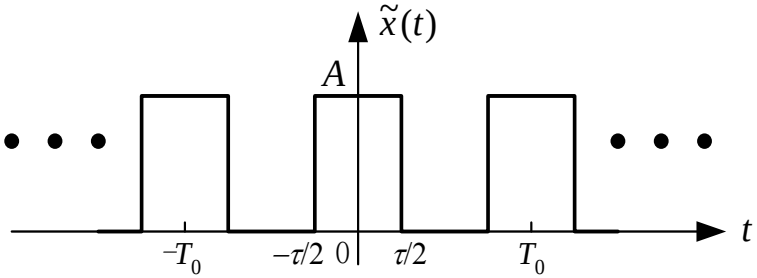
$$= \frac{1}{T_0}\int_{-\frac{\tau}{2}}^{\frac{\tau}{2}} A\mathrm{e}^{-\mathrm{j}n\omega_0 t}\mathrm{d}t$$

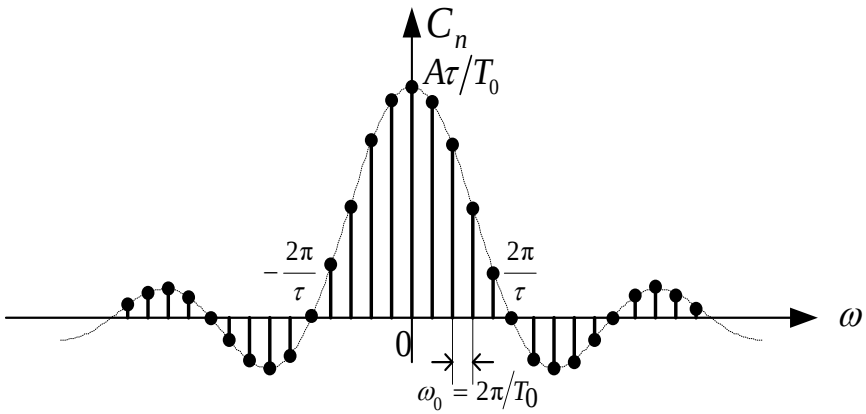
$$= \frac{A\tau}{T_0}\mathrm{Sa}\left(\frac{n\omega_0\tau}{2}\right)$$

Spectrum of a periodic rectangular signal



Ex. 1 Try to calculate the periodic rectangular pulse signal $\tilde{x}(t)$ shown in the figure.





Solution:

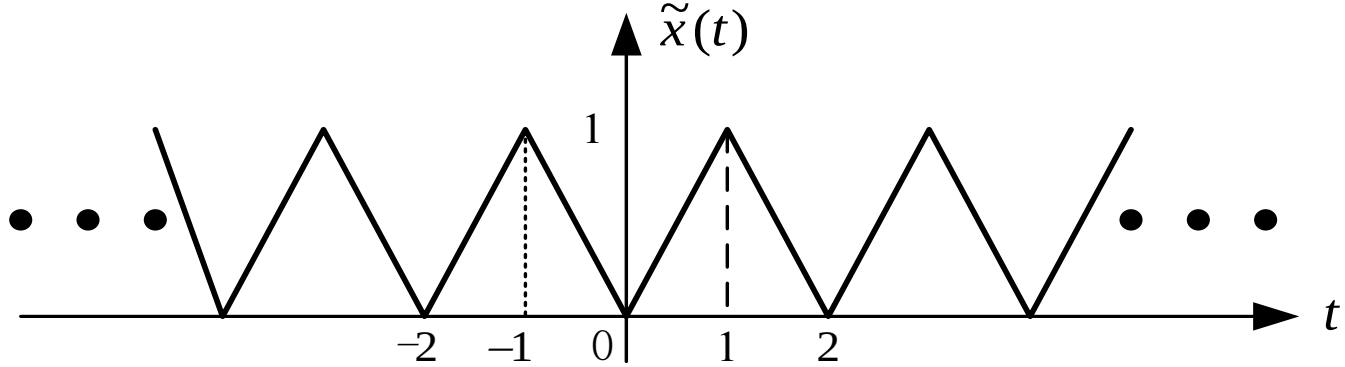
Exponential form $$\tilde{x}(t) = \sum_{n=-\infty}^{\infty} C_n \, \mathrm{e}^{\mathrm{j}n\omega_0 t} = \frac{A\tau}{T_0} \sum_{n=-\infty}^{\infty} \mathrm{Sa}\left(\frac{n\omega_0\tau}{2}\right) \mathrm{e}^{\mathrm{j}n\omega_0 t} \qquad \omega_0 = \frac{2\pi}{T_0}$$

Triangular form $$= \frac{\tau A}{T_0} + \sum_{n=1}^{\infty} \frac{2\tau A}{T_0} \mathrm{Sa}\left(\frac{n\omega_0\tau}{2}\right) \cos(n\omega_0 t)$$

If $\tau = T_0/2$, then

$$\tilde{x}(t) = \frac{A}{2} + \frac{2A}{\pi}\left(\cos\omega_0 t - \frac{1}{3}\cos 3\omega_0 t + \frac{1}{5}\cos 5\omega_0 t - \cdots\right) \qquad \omega_0 = \frac{2\pi}{T_0}$$

Ex. 2 Try to calculate the spectrum of the illustrated periodic triangular pulse signal and write its Fourier series expression.



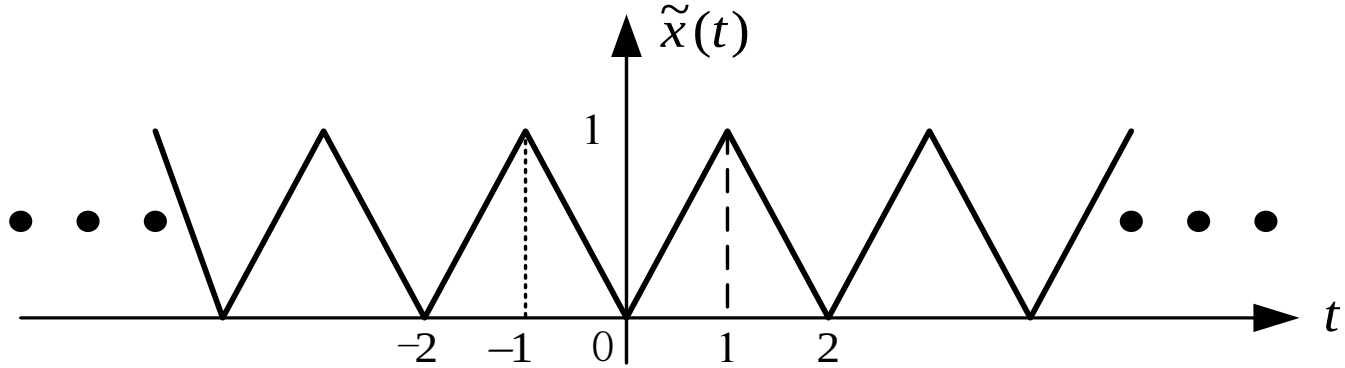
Solution:

$$C_n = \frac{1}{T_0}\int_{-T_0/2}^{T_0/2}\tilde{x}(t)\mathrm{e}^{-\mathrm{j}n\omega_0 t}\mathrm{d}t = \frac{1}{2}\left(\int_{-1}^{0} -t\mathrm{e}^{-\mathrm{j}n\omega_0 t}\mathrm{d}t + \int_{0}^{1} t\mathrm{e}^{-\mathrm{j}n\omega_0 t}\mathrm{d}t\right)$$

$$= \frac{1}{2\mathrm{j}n\omega_0}\left(t\mathrm{e}^{-\mathrm{j}n\omega_0 t}\Big|_{-1}^{0} - \int_{-1}^{0}\mathrm{e}^{-\mathrm{j}n\omega_0 t}\mathrm{d}t - t\mathrm{e}^{-\mathrm{j}n\omega_0 t}\Big|_{0}^{1} + \int_{0}^{1}\mathrm{e}^{-\mathrm{j}n\omega_0 t}\mathrm{d}t\right)$$

$$= \frac{1}{(n\pi)^2}(\cos n\pi - 1) = \begin{cases} -2/(n\pi)^2, n\text{为奇数} \\ 1/2, \qquad n = 0 \end{cases} \qquad \omega_0 = \frac{2\pi}{T_0} = \pi$$

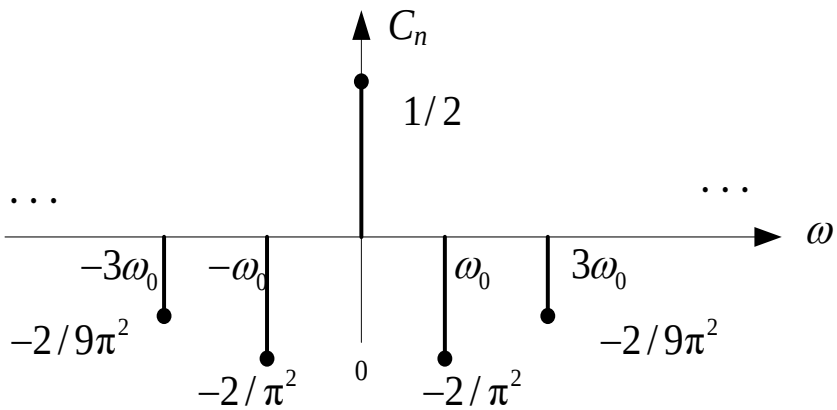
Ex. 2 Try to calculate the spectrum of the illustrated periodic triangular pulse signal and write its Fourier series expression.



Solution:

$$C_n = \frac{1}{(n\pi)^2}(\cos n\pi - 1)$$

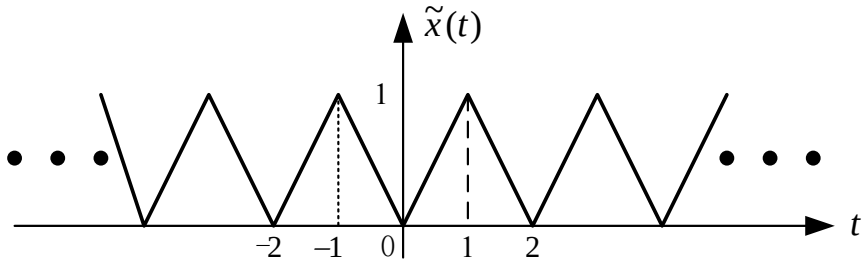
$$= \begin{cases} -2/(n\pi)^2, n \text{ is odd} \\ 1/2, \qquad n = 0 \end{cases}$$

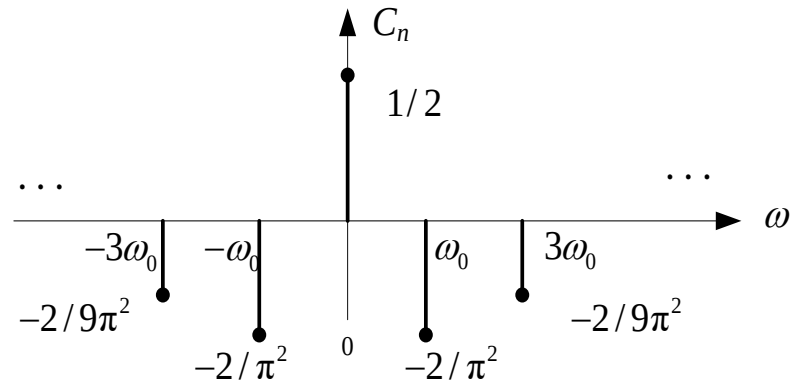


Spectrum of periodic triangular wave signals

$$\omega_0 = \frac{2\pi}{T_0} = \pi$$

Ex. 2 Try to calculate the spectrum of the illustrated periodic triangular pulse signal and write its Fourier series expression.





Solution:

Exponential form $$\tilde{x}(t)=\sum_{n=-\infty}^{\infty}C_n\,\mathrm{e}^{\mathrm{j}n\omega_0 t}=\frac{1}{2}-\sum_{m=-\infty}^{\infty}\frac{2}{[(2m-1)\pi]^2}\mathrm{e}^{\mathrm{j}(2m-1)\omega_0 t}$$

Triangular form $$=\frac{1}{2}-\sum_{m=1}^{\infty}\frac{4}{[(2m-1)\pi]^2}\cos[(2m-1)\omega_0 t] \qquad \omega_0=\frac{2\pi}{T_0}=\pi$$

$$=\frac{1}{2}-\frac{4}{\pi^2}\cos\omega_0 t-\frac{4}{9\pi^2}\cos 3\omega_0 t-\frac{4}{25\pi^2}\cos 5\omega_0 t-\cdots$$

Spectrum of Continuous Periodic Signals

Periodic signals can be decomposed into the sum of imaginary exponential signals of different frequencies.

$$\tilde{x}(t) = \sum_{n=-\infty}^{\infty} C_n \, \mathrm{e}^{\mathrm{j}n\omega_0 t}$$

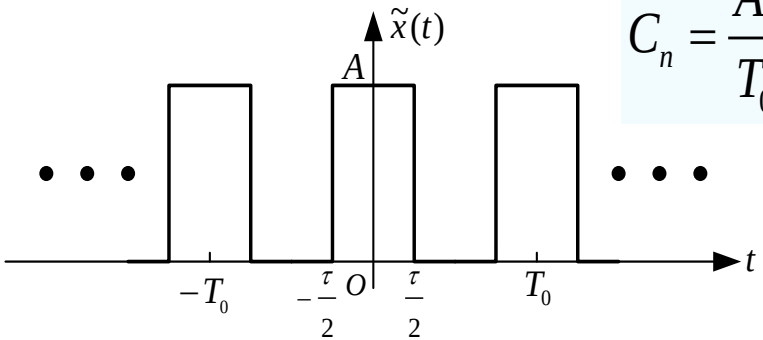
eg. The following signals

$$\tilde{x}_{\mathrm{r}}(t) = \frac{A}{2} + \frac{2A}{\pi}\cos\omega_0 t - \frac{2A}{3\pi}\cos 3\omega_0 t + \frac{2A}{5\pi}\cos 5\omega_0 t - \cdots$$

$$\tilde{x}_{\Delta}(t) = \frac{1}{2} - \frac{4}{\pi^2}\cos\omega_0 t - \frac{4}{9\pi^2}\cos 3\omega_0 t - \frac{4}{25\pi^2}\cos 5\omega_0 t - \cdots$$

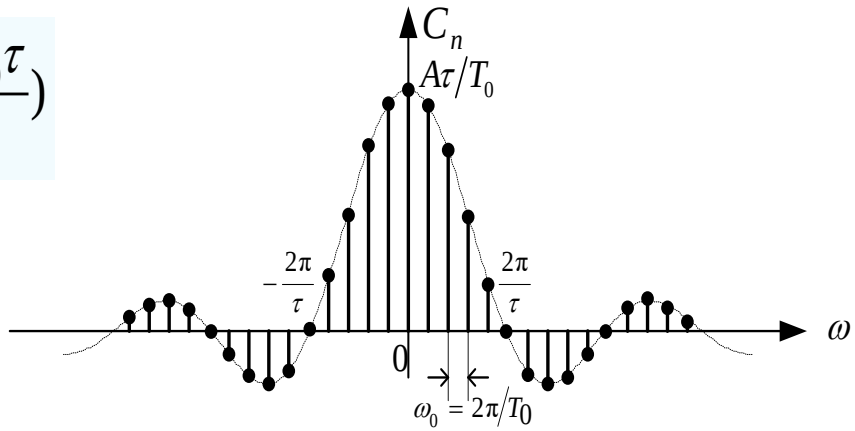
Different time-domain signals differ only in the Fourier series coefficients C_n, hence by studying the Fourier series coefficients, we can investigate the characteristics of the signals.

Spectrum of Continuous Periodic Signals



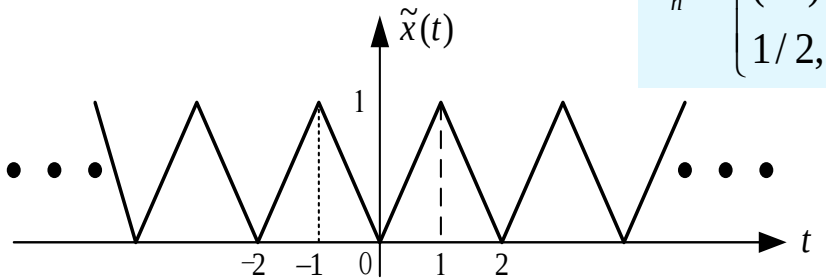
$$C_n = \frac{A\tau}{T_0}\,\mathrm{Sa}\left(\frac{n\omega_0\tau}{2}\right)$$

Time-domain waveform of periodic rectangular signals

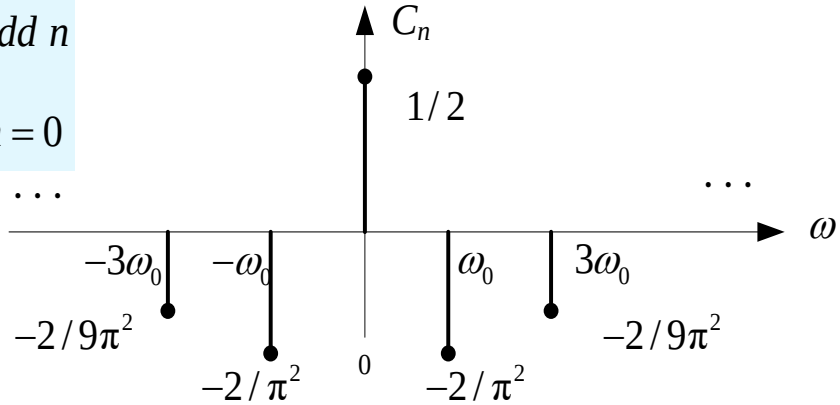


Spectrum of periodic rectangular signals

$$C_n = \begin{cases} \dfrac{-2}{(n\pi)^2}, & \text{for odd } n \\ 1/2, & n = 0 \end{cases}$$



Time-domain waveform of periodic triangular signals



Spectrum of periodic triangular signals

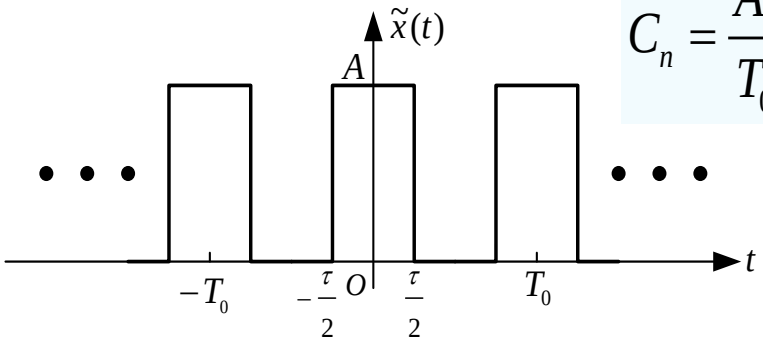
Spectrum of Continuous Periodic Signals

Periodic signals can be decomposed into the sum of sinusoidal (or exponential) signals of different frequencies.

- From the perspective of signal analysis, representing signals as a linear combination of sinusoidal components of different frequencies provides a means for comparing different signals.

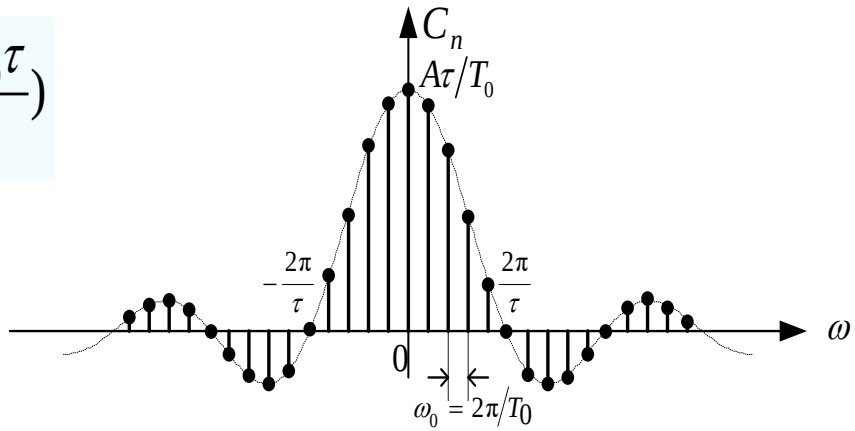
- From the perspective of system analysis, knowing the response to a single-frequency sinusoidal signal, the principle of superposition can be used to determine the total response to multiple sinusoidal signals of different frequencies acting simultaneously, as well as the changes each sinusoidal component undergoes as it passes through the system.

Spectrum of Continuous Periodic Signals



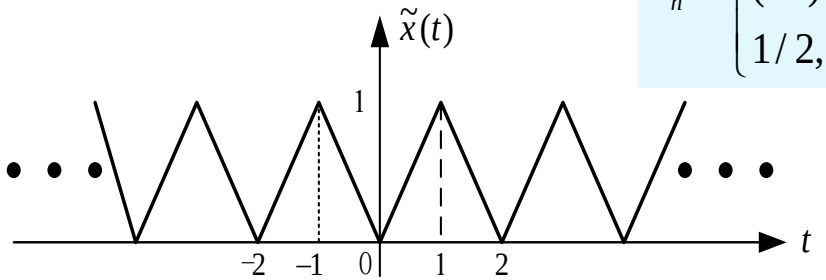
$$C_n = \frac{A\tau}{T_0}\,\mathrm{Sa}\left(\frac{n\omega_0\tau}{2}\right)$$

Time-domain waveform of periodic rectangular signals

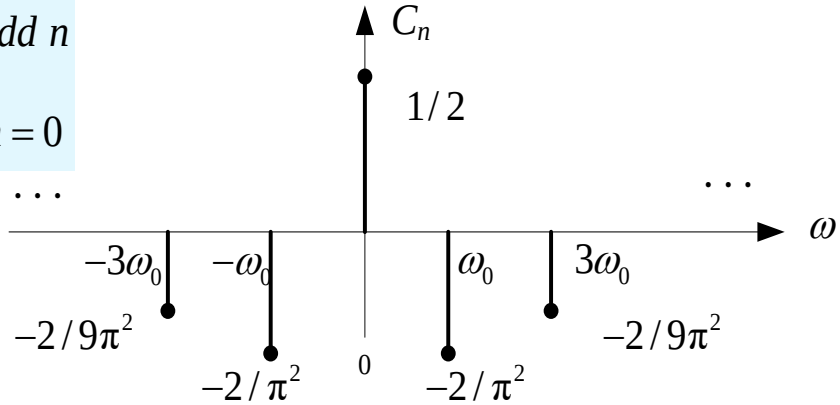


Spectrum of periodic rectangular signals

$$C_n = \begin{cases} \dfrac{-2}{(n\pi)^2}, & \text{for odd } n \\ 1/2, & n = 0 \end{cases}$$



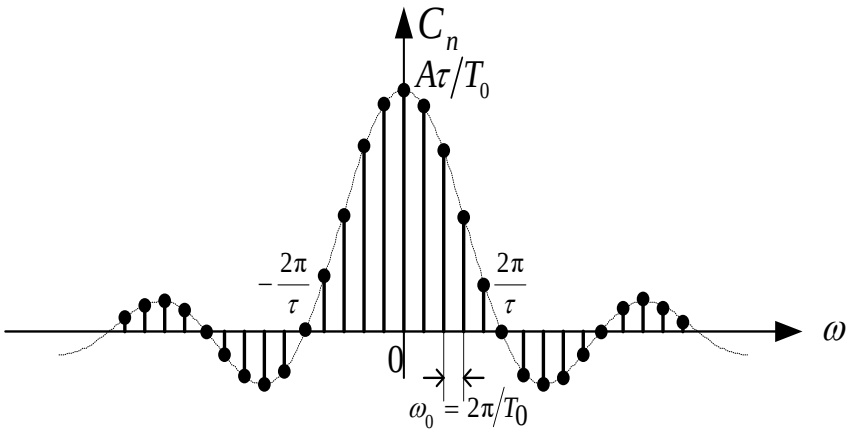
Time-domain waveform of periodic triangular signals



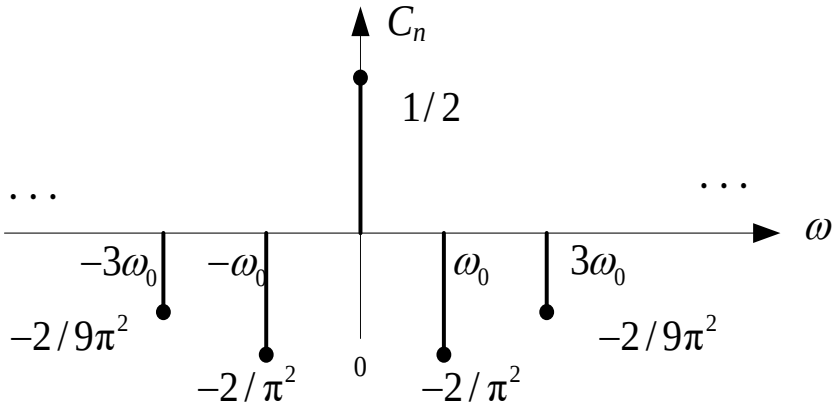
Spectrum of periodic triangular signals

Spectrum Characteristics of Continuous Periodic Signals

- **Dispersion Characteristic:** The spectrum of a periodic signal is composed of spectral lines spaced by ω_0.



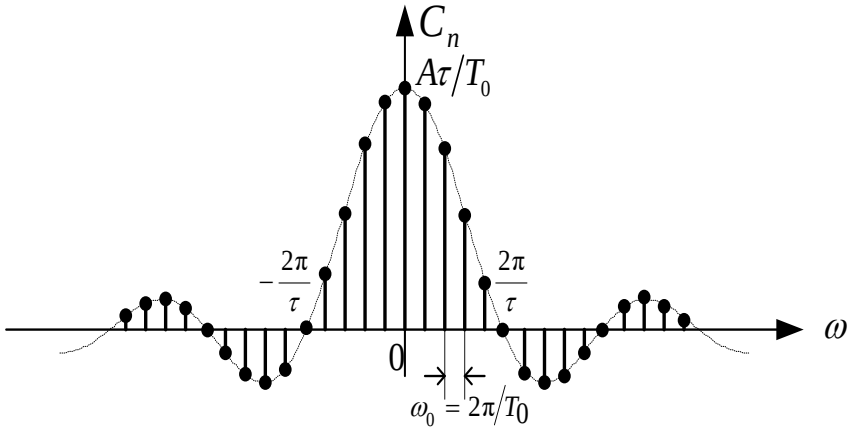
Spectrum of Periodic Rectangular Signals



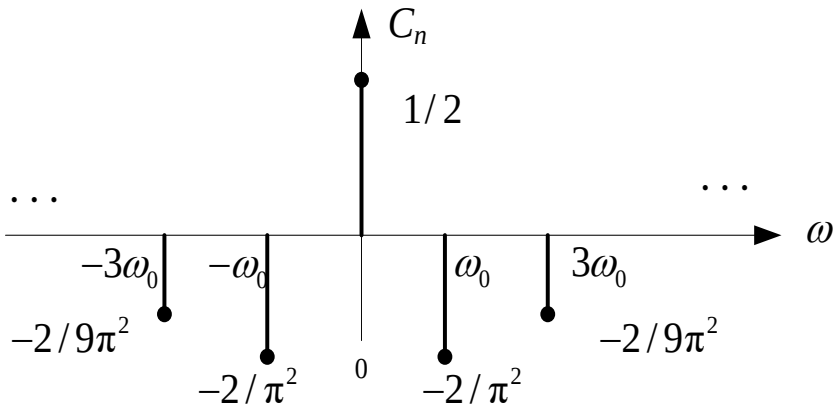
Spectrum of Periodic Triangular Signals

Spectrum Characteristics of Continuous Periodic Signals

- **Decay Characteristic:** The amplitude spectrum $|C_n|$ gradually decreases as the harmonic number $n\omega_0$ increases, and eventually tends towards zero.



Spectrum of Periodic Rectangular Signals

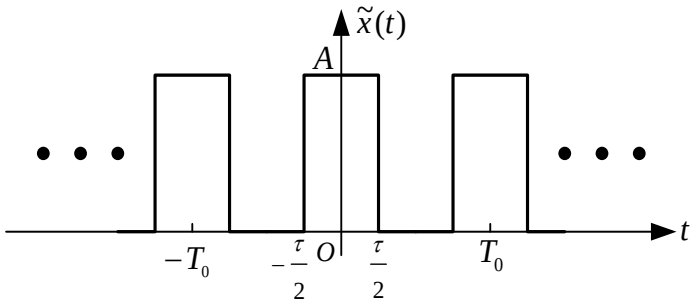


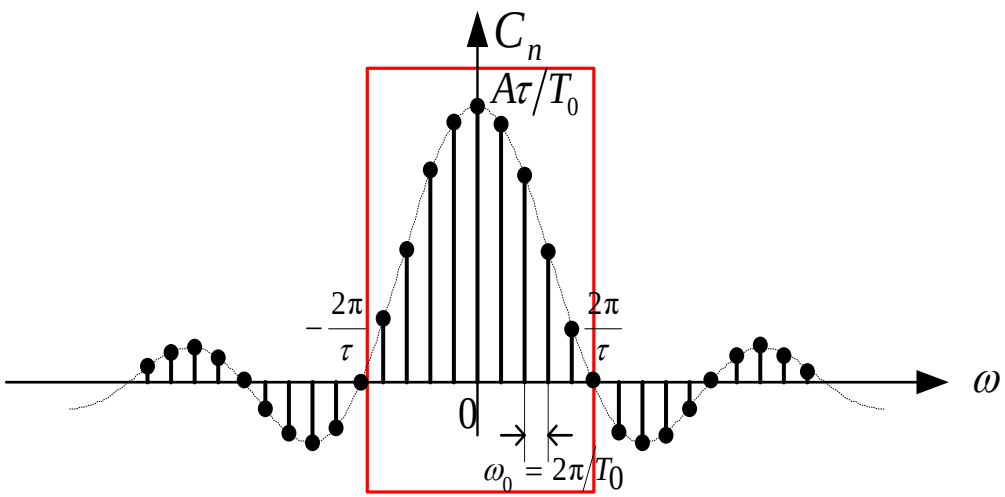
Spectrum of Periodic Triangular Signals

The smoother the change in the time-domain waveform of the signal, the fewer the high-order harmonic components, and the faster the amplitude spectrum decays; the more abrupt the changes in the time-domain waveform, the more high-order harmonic components there are, and the slower the amplitude spectrum decays.

Spectrum Characteristics of Continuous Periodic Signals

Effective Bandwidth



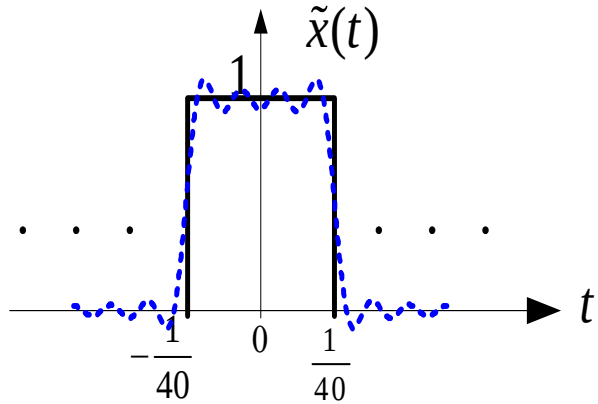


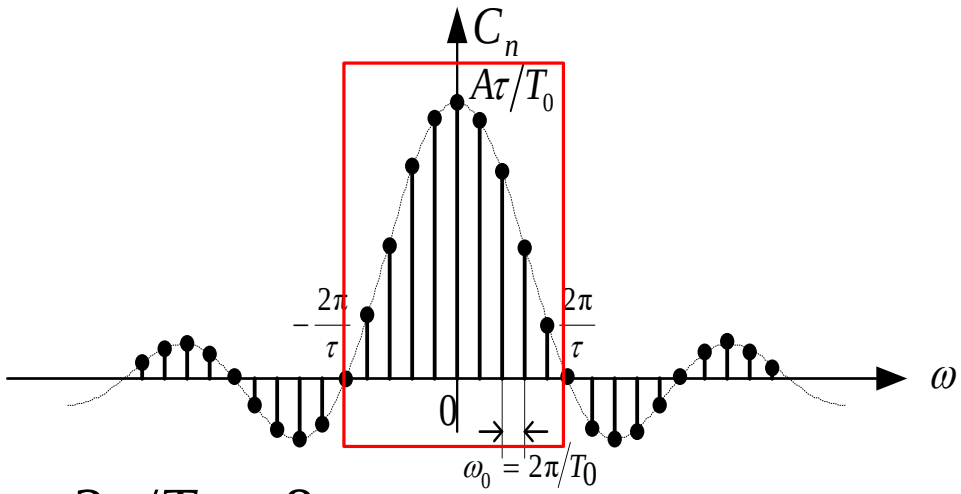
The frequency range $(0 \sim \frac{2\pi}{\tau})$ that usually contains the main harmonic components is referred to as the effective bandwidth $\omega_B = \frac{2\pi}{\tau}$ of the periodic rectangular signal.

The effective bandwidth of the signal is inversely proportional to the duration in the time domain.

Spectrum Characteristics of Continuous Periodic Signals

- The harmonic components outside the effective bandwidth will not significantly affect the signal





$A=1$, $T_0=1/4$, $\tau = 1/20$, $\omega_0 = 2\pi/T_0 = 8\pi$

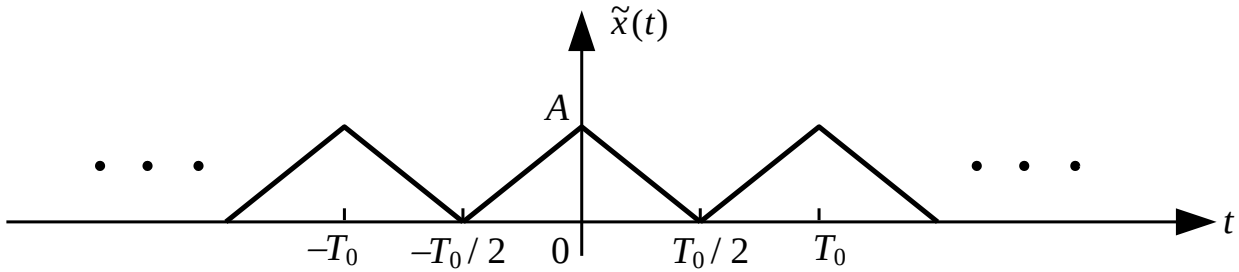
$$C_n = 0.2\,\mathrm{Sa}(n\omega_0/40) = 0.2\,\mathrm{Sa}(n\pi/5)$$

The first zero point appears at $\dfrac{2\pi}{\tau} = 40\pi = 5\times 8\pi = 5\omega_0$

Spectrum Characteristics of Continuous Periodic Signals

◆Symmetry Characteristics

(1) Vertically Symmetric Signal $\tilde{x}(t)=\tilde{x}(-t)$



$$a_n=\frac{2}{T_0}\int_{-T_0/2}^{T_0/2}\tilde{x}(t)\cos(n\omega_0 t)\mathrm{d}t=\frac{4}{T_0}\int_{0}^{T_0/2}\tilde{x}(t)\cos(n\omega_0 t)\mathrm{d}t$$

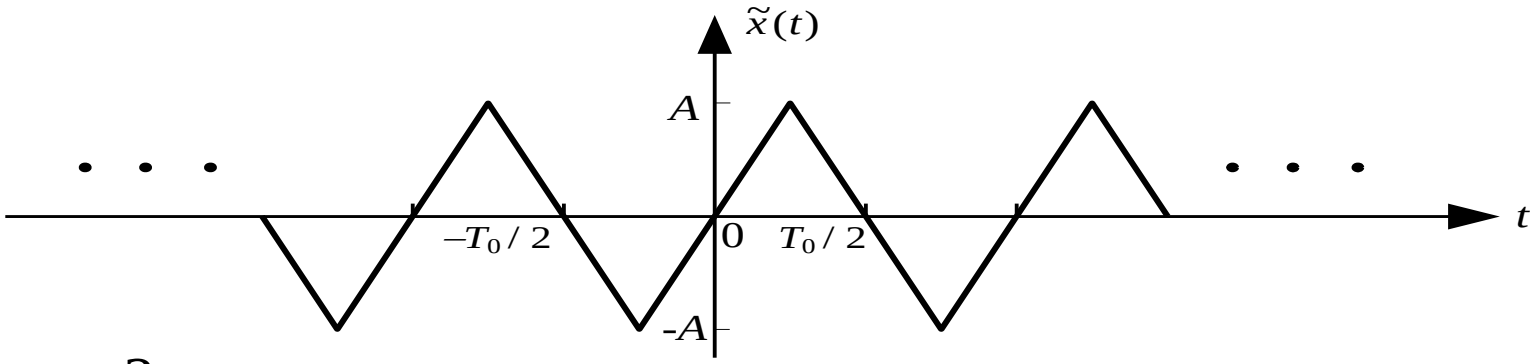
$$b_n=\frac{2}{T_0}\int_{-T_0/2}^{T_0/2}\tilde{x}(t)\sin(n\omega_0 t)\mathrm{d}t=0$$

The Fourier series expansion of a vertically symmetric periodic signal contains only the DC component and cosine terms.

Spectrum Characteristics of Continuous Periodic Signals

◆Symmetry Characteristics

(2) Origin-Symmetric Signal $\tilde{x}(t) = -\tilde{x}(-t)$



$$a_n = \frac{2}{T_0}\int_{-T_0/2}^{T_0/2} \tilde{x}(t)\cos(n\omega_0 t)\mathrm{d}t = 0$$

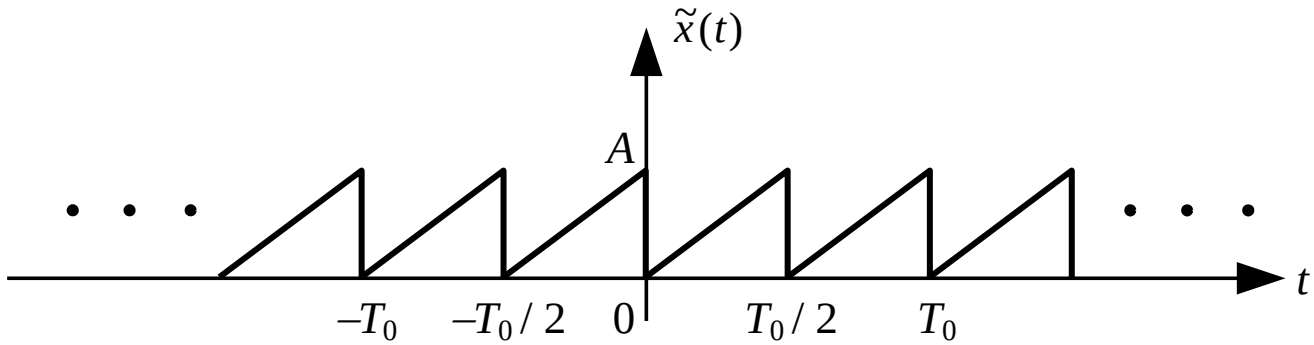
$$b_n = \frac{2}{T_0}\int_{-T_0/2}^{T_0/2} \tilde{x}(t)\sin(n\omega_0 t)\mathrm{d}t = \frac{4}{T_0}\int_{0}^{T_0/2} \tilde{x}(t)\sin(n\omega_0 t)\mathrm{d}t$$

The Fourier series expansion of an origin-symmetric periodic signal contains only sine terms.

Spectrum Characteristics of Continuous Periodic Signals

◆Symmetry Characteristics

(3) Half-Wave Symmetry Signal $\tilde{x}(t) = \tilde{x}(t \pm T_0 / 2)$

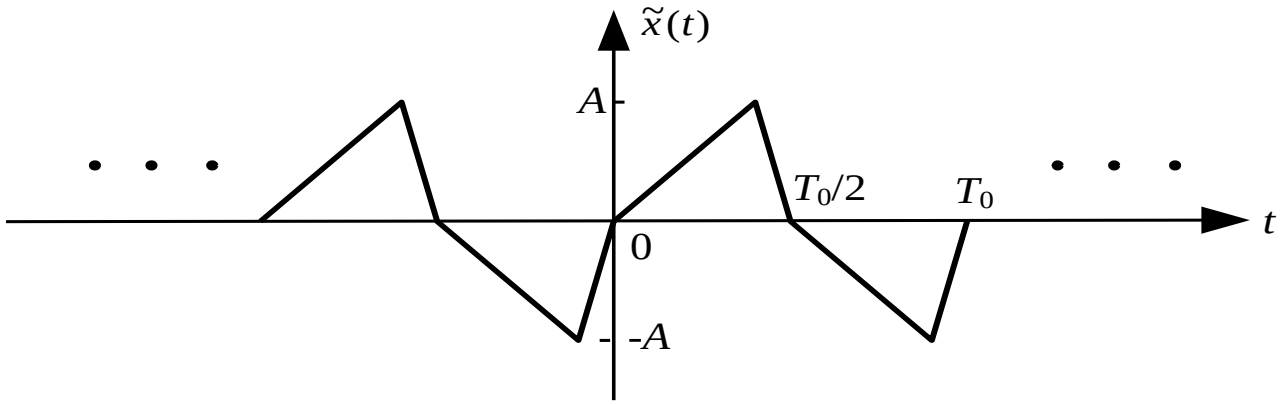


Half-wave symmetric periodic signals contain only even harmonics of sine and cosine, with no odd harmonic components.

Spectrum Characteristics of Continuous Periodic Signals

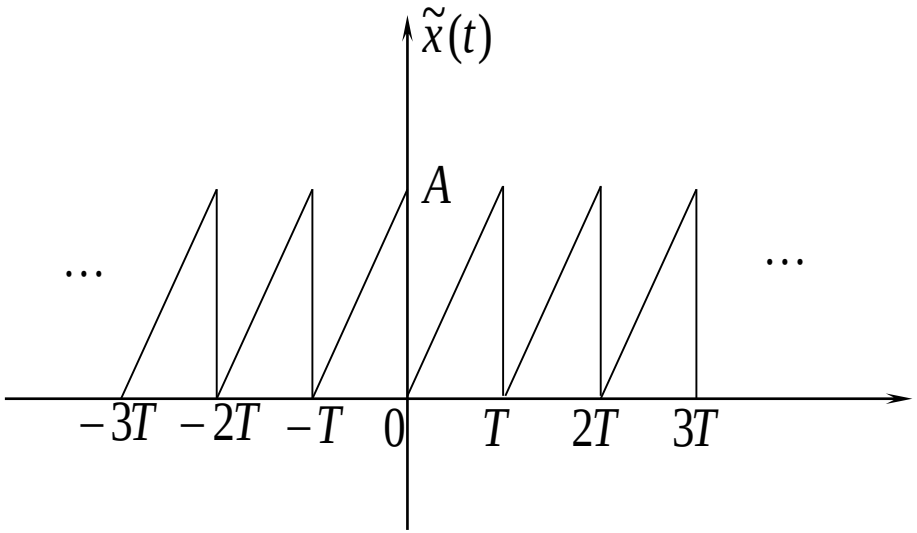
◆Symmetry Characteristics

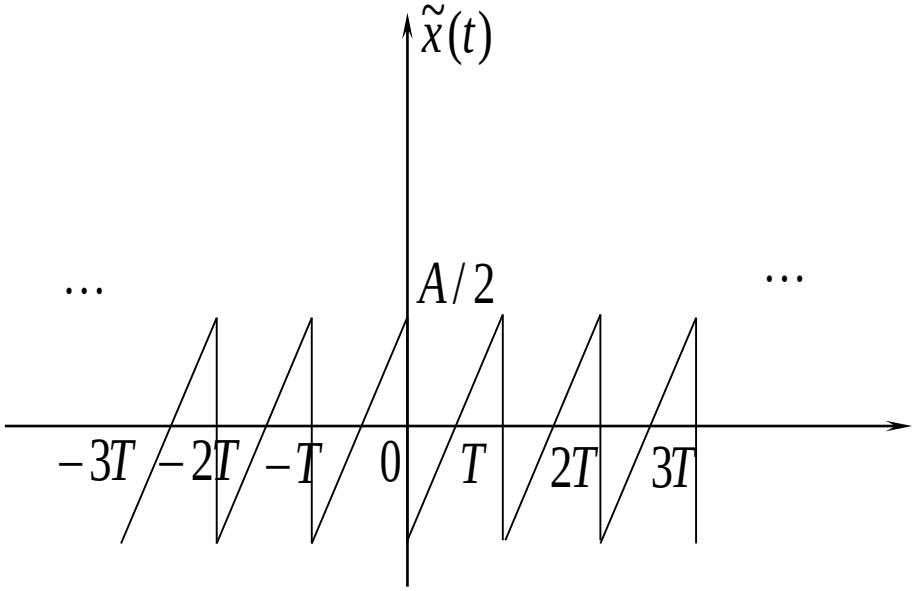
(4) Half-Wave Image Signal $\tilde{x}(t) = -\tilde{x}(t \pm T_0 / 2)$



The Fourier series expansion of a half-wave image periodic signal contains only odd harmonics of sine and cosine, with no DC component or even harmonics.

Note: Some signal waveforms only exhibit certain symmetry characteristics after being shifted up, down, left, or right.



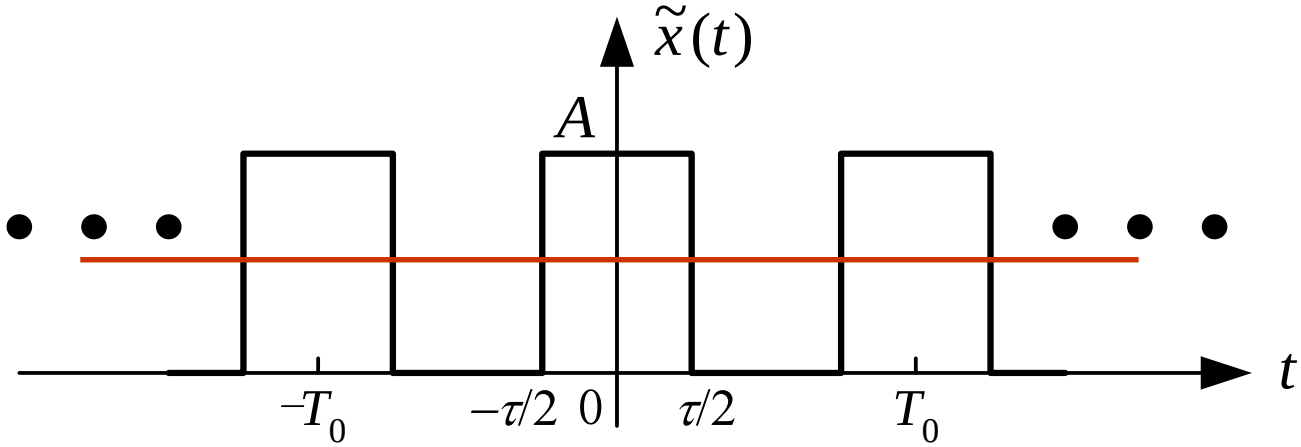


After removing the DC component,

The signal is odd-symmetric, containing only sine harmonic components.

Therefore, the signal contains sine harmonic components and a DC component.

Ex. 1: Frequency Spectrum Components of a Periodic Rectangular Pulse Signal



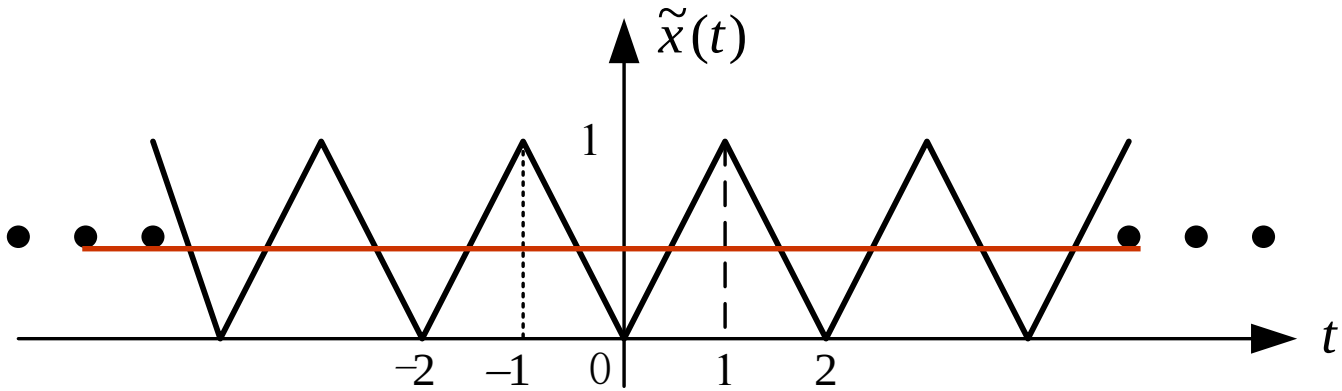
Solution: If $\tau = T_0/2$, we have

$$\tilde{x}(t) = \frac{A}{2} + \frac{2A}{\pi}\left(\cos\omega_0 t - \frac{1}{3}\cos 3\omega_0 t + \frac{1}{5}\cos 5\omega_0 t - \cdots\right) \quad \omega_0 = \frac{2\pi}{T_0}$$

The signal is even-symmetric, after removing the DC component, it becomes a half-wave mirror signal,

hence the signal contains the DC component and the cosine components of odd harmonics.

Ex. 2: Frequency Spectrum Components of a Periodic Triangular Pulse Signal



Solution:

$$\tilde{x}_{\Delta}(t) = \frac{1}{2} - \frac{4}{\pi^2}\cos\omega_0 t - \frac{4}{9\pi^2}\cos 3\omega_0 t - \frac{4}{25\pi^2}\cos 5\omega_0 t - \cdots$$

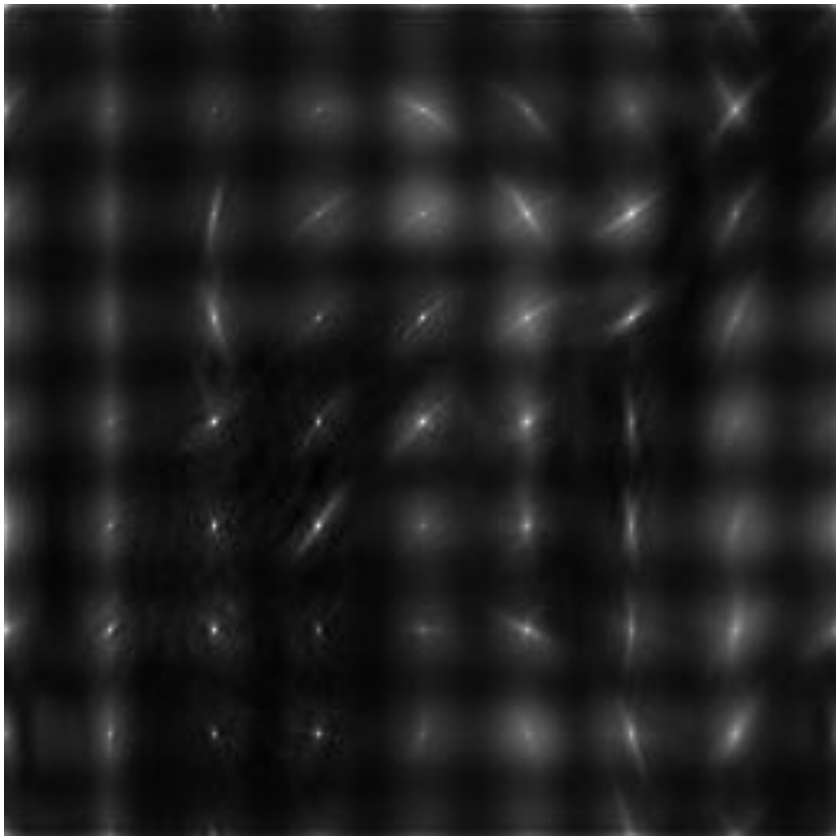
$$\omega_0 = \frac{2\pi}{T_0} = \pi$$

The signal is even-symmetric, after removing the DC component, it becomes a half-wave mirror signal,

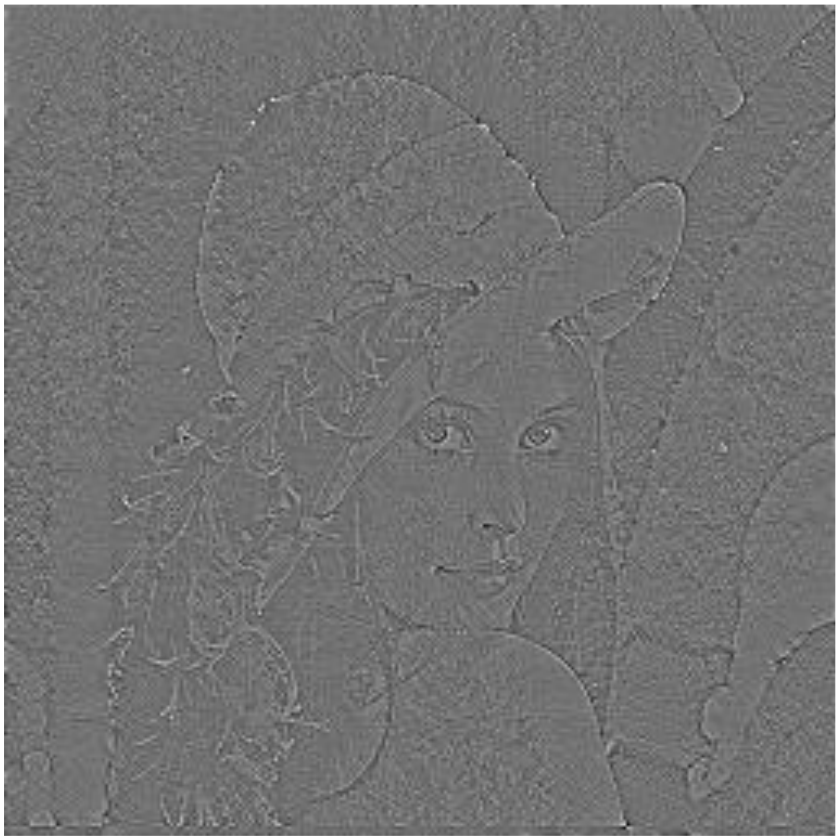
hence the signal contains the DC component and the cosine components of odd harmonics.

Characteristics of the Spectrum of Continuous Periodic Signals

◆The Role of Phase Spectrum

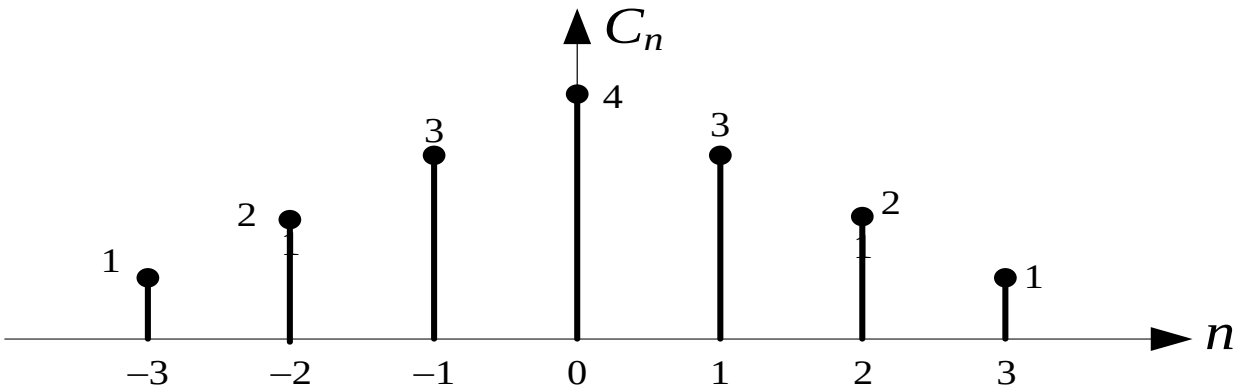


Constant amplitude and frequency, zero phase



Constant amplitude, unchanging phase

Ex. 3: Given the spectrum of the continuous periodic signal $x_1(t)$ as shown in the figure, write the Fourier series representation of the signal.



Solution: From the diagram, it can be seen that

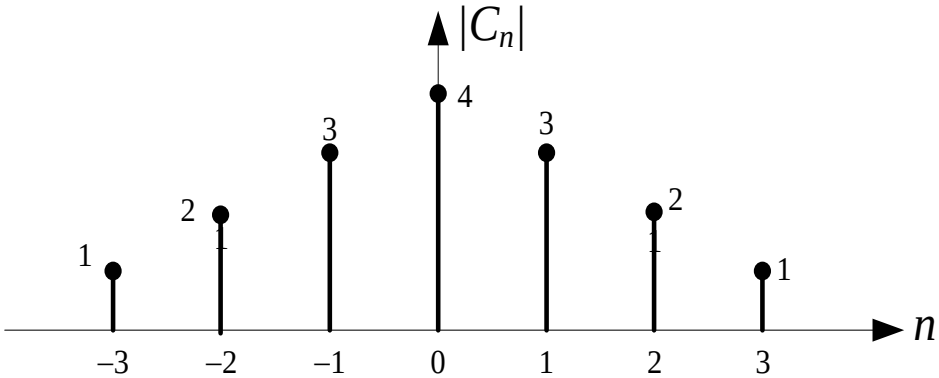
$$C_0 = 4 \qquad C_{\pm 1} = 3 \qquad C_{\pm 2} = 2 \qquad C_{\pm 3} = 1$$

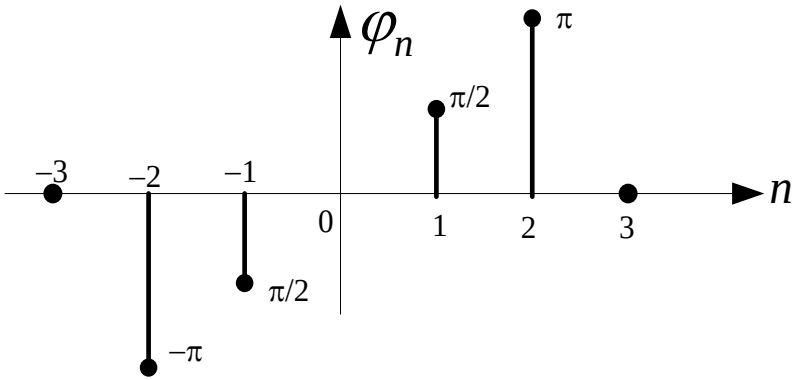
$$\tilde{x}(t) = \sum_{n=-\infty}^{\infty} C_n \mathrm{e}^{\mathrm{j}n\omega_0 t}$$

$$= 4 + 3(\mathrm{e}^{\mathrm{j}\omega_0 t} + \mathrm{e}^{-\mathrm{j}\omega_0 t}) + 2(\mathrm{e}^{\mathrm{j}2\omega_0 t} + \mathrm{e}^{-\mathrm{j}2\omega_0 t}) + (\mathrm{e}^{\mathrm{j}3\omega_0 t} + \mathrm{e}^{-\mathrm{j}3\omega_0 t})$$

$$= 4 + 6\cos(\omega_0 t) + 4\cos(2\omega_0 t) + 2\cos(3\omega_0 t)$$

Ex. 4: Given the spectrum of the continuous periodic signal $x_2(t)$ as shown in the figure, write the Fourier series representation of the signal.





Solution: From the diagram, it can be seen that

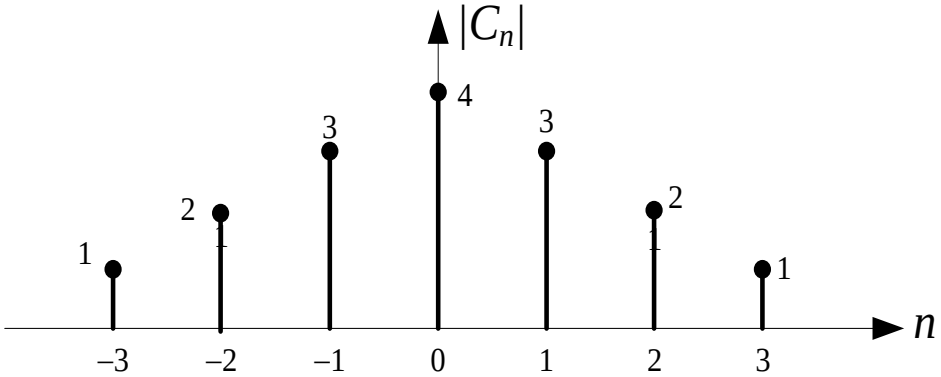
$$C_0 = 4 \quad C_{\pm 1} = 3\mathrm{e}^{\pm \mathrm{j}\frac{\pi}{2}} \quad C_{\pm 2} = 2\mathrm{e}^{\pm \mathrm{j}\pi} \quad C_{\pm 3} = 1$$

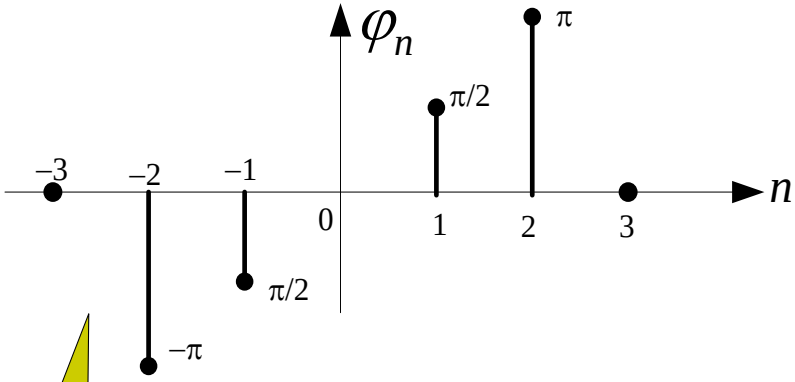
$$\tilde{x}(t) = \sum_{n=-\infty}^{\infty} C_n \mathrm{e}^{\mathrm{j}n\omega_0 t} = 4 + 3(\mathrm{e}^{\mathrm{j}(\omega_0 t + \pi/2)} + \mathrm{e}^{-\mathrm{j}(\omega_0 t + \pi/2)})$$

$$+ 2(\mathrm{e}^{\mathrm{j}(2\omega_0 t + \pi)} + \mathrm{e}^{-\mathrm{j}(2\omega_0 t + \pi)}) + (\mathrm{e}^{\mathrm{j}3\omega_0 t} + \mathrm{e}^{-\mathrm{j}3\omega_0 t})$$

$$= 4 - 6\sin(\omega_0 t) - 4\cos(2\omega_0 t) + 2\cos(3\omega_0 t)$$

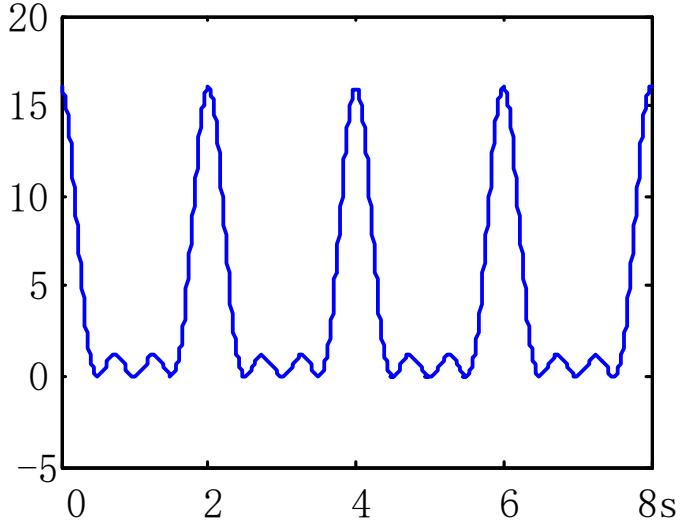
Time-domain Signal Waveforms Corresponding to the Spectrum of Ex. 3 and Ex. 4

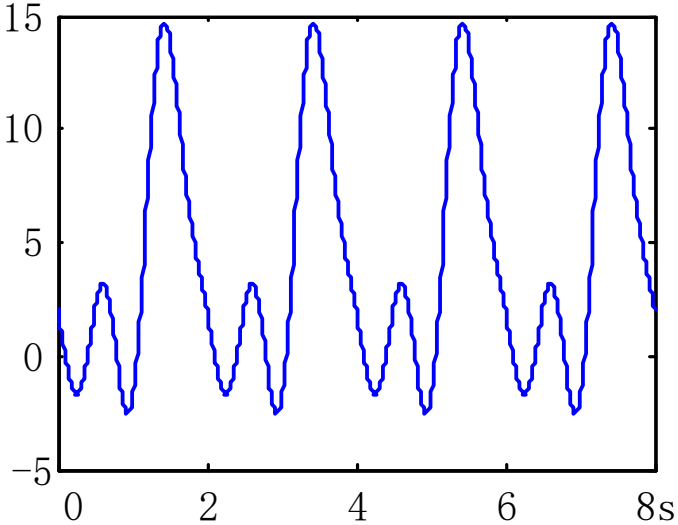




$$\tilde{x}_1(t) = 4 + 6\cos(\omega_0 t) + 4\cos(2\omega_0 t) + 2\cos(3\omega_0 t)$$

$$\tilde{x}_2(t) = 4 - 6\sin(\omega_0 t) - 4\cos(2\omega_0 t) + 2\cos(3\omega_0 t)$$





Basic Properties of Fourier Series

- Linearity

$$\text{If} \quad \tilde{x}_1(t) \to C_{1n}\ , \quad \tilde{x}_2(t) \to C_{2n}$$

$$\text{Then} \quad a_1 \cdot \tilde{x}_1(t) + a_2 \cdot \tilde{x}_2(t) \to a_1 \cdot C_{1n} + a_2 \cdot C_{2n}$$

- Symmetry

If $\tilde{x}(t)$ is a real signal,

$$\text{Then} \quad |C_n| = |C_{-n}| \qquad \varphi_n = -\varphi_{-n}$$

Basic Properties of Fourier Series

- Time-shifting Property

$$\text{If} \qquad \tilde{x}(t) \rightarrow C_n$$

$$\text{Then} \qquad \tilde{x}(t-t_0) \rightarrow \mathrm{e}^{-\mathrm{j}n\omega_0 t_0} C_n$$

- Frequency-shifting Property

$$\text{If} \qquad \tilde{x}(t) \rightarrow C_n$$

$$\textit{Then} \qquad \tilde{x}(t)\mathrm{e}^{\mathrm{j}M\omega_0 t} \longleftrightarrow C_{n-M}$$

Basic Properties of Fourier Series

Convolution Property

If $\tilde{x}_1(t)$ *and* $\tilde{x}_2(t)$ are both periodic signals with period T_0,

$$\tilde{x}_1(t) \rightarrow C_{1n}\ , \qquad \tilde{x}_2(t) \rightarrow C_{2n}$$

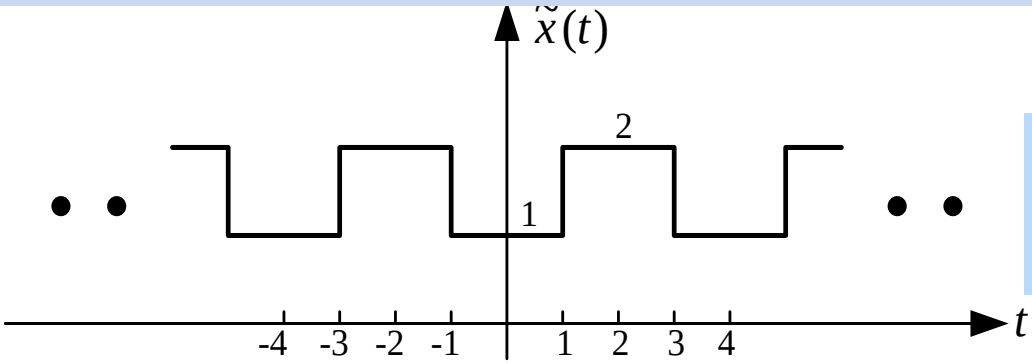
Then $\quad \tilde{x}_1(t) * \tilde{x}_2(t) \rightarrow T_0 C_{1n} \cdot C_{2n}$

Differentiation Property

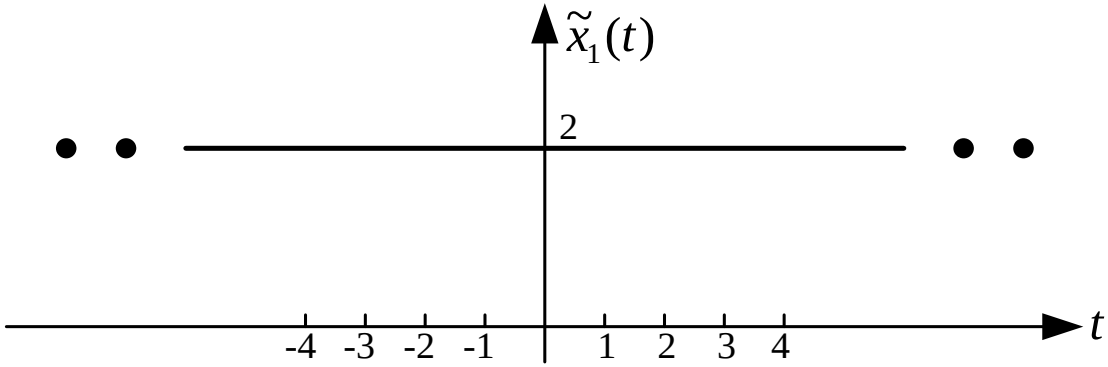
If $\quad \tilde{x}(t) \rightarrow C_n$

Then $\quad \tilde{x}'(t) \rightarrow \mathrm{j}n\omega_0 C_n$

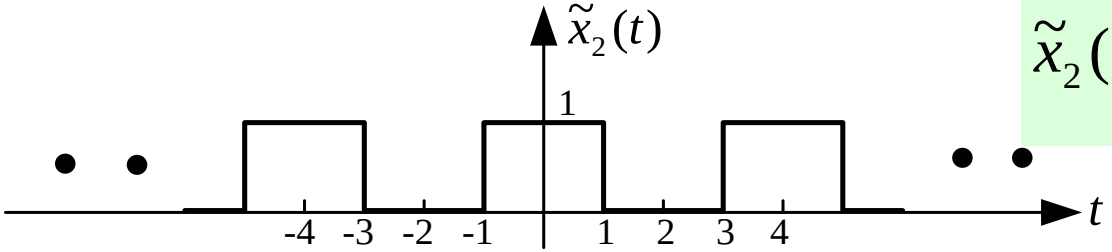
Ex. 5: Find the Fourier Series of the Given Periodic Signal



$$\tilde{x}(t) = 1.5 - \sum_{n=1}^{\infty} \mathrm{Sa}(\frac{n\pi}{2})\cos(\frac{n\pi t}{2})$$



$$\tilde{x}(t) = \tilde{x}_1(t) - \tilde{x}_2(t)$$



$$\tilde{x}_2(t) = 0.5 + \sum_{n=1}^{\infty} \mathrm{Sa}(\frac{n\pi}{2})\cos(\frac{n\pi t}{2})$$

Power Spectrum of Periodic Signals

Parseval Power Conservation Theorem

$$P = \frac{1}{T_0}\int_{-T_0/2}^{T_0/2} |\tilde{x}(t)|^2 \, \mathrm{d}t = \sum_{n=-\infty}^{\infty} |C_n|^2$$

Physical significance: The average power of any periodic signal is equal to the sum of the average powers of the DC component, fundamental wave, and each harmonic component contained in the signal.

The power spectrum of a periodic signal: The distribution of $|C_n|^2$ with respect to nw_0 is referred to as the power spectrum of the periodic signal, also known as the power spectrum.

Ex. 6 If $\tilde{x}(t)=2\mathrm{e}^{-\mathrm{j}2\omega_0 t}+3\mathrm{e}^{-\mathrm{j}\omega_0 t}+4+3\mathrm{e}^{\mathrm{j}\omega_0 t}+2\mathrm{e}^{\mathrm{j}2\omega_0 t}$, find the power

Solution:

(1) According to Parseval Power Conservation Theorem

$$P=\frac{1}{T_0}\int_{-T_0/2}^{T_0/2}\left|\tilde{x}(t)\right|^2\mathrm{d}t=\sum_{n=-\infty}^{\infty}\left|C_n\right|^2$$

$$\tilde{x}(t)=2\mathrm{e}^{-\mathrm{j}2\omega_0 t}+3\mathrm{e}^{-\mathrm{j}\omega_0 t}+4+3\mathrm{e}^{\mathrm{j}\omega_0 t}+2\mathrm{e}^{\mathrm{j}2\omega_0 t}$$

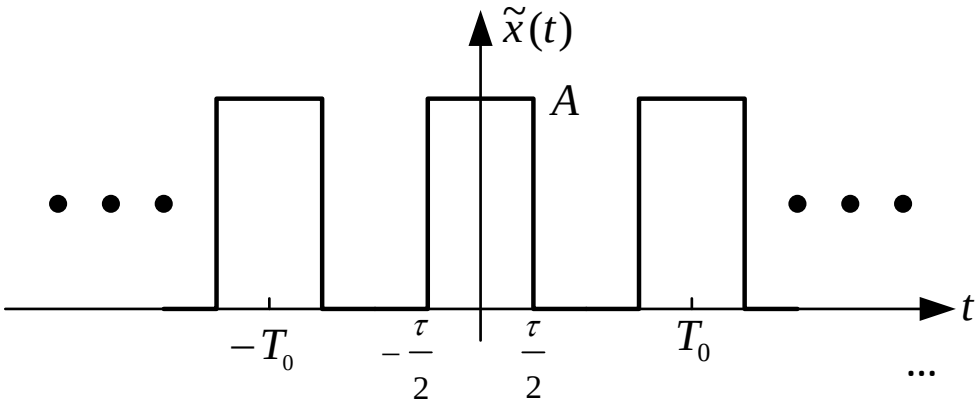
$$C_{-2}=2 \quad C_{-1}=3 \quad C_0=4 \quad C_1=3 \quad C_2=2$$

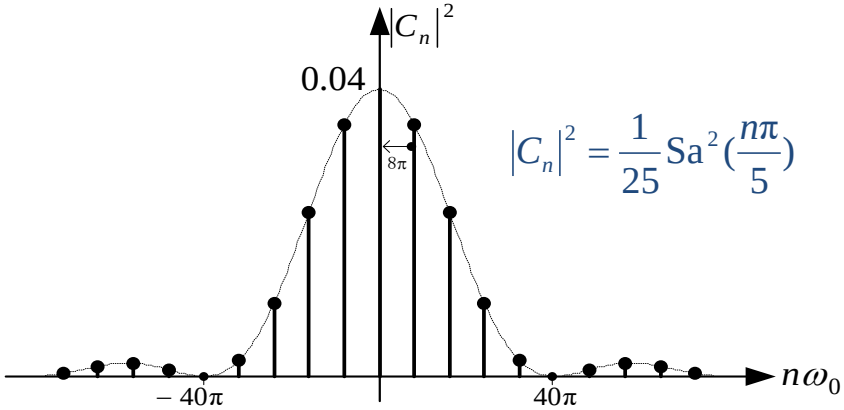
$$P=2^2+3^2+4^2+3^2+2^2=42$$

(2) $$\tilde{x}(t)=4+6\cos\omega_0 t+4\cos 2\omega_0 t$$

$$P=4^2+\frac{1}{2}\times 6^2+\frac{1}{2}\times 4^2=42$$

Ex.7 : Calculate the percentage of the average power of the harmonic components within the effective bandwidth ($0\sim 2\pi/\tau$) of a periodic rectangular pulse signal relative to the average power of the entire signal. Given A=1，T_0=1/4，τ=1/20.





Solution: The Fourier coefficients of the periodic rectangular pulse are given by

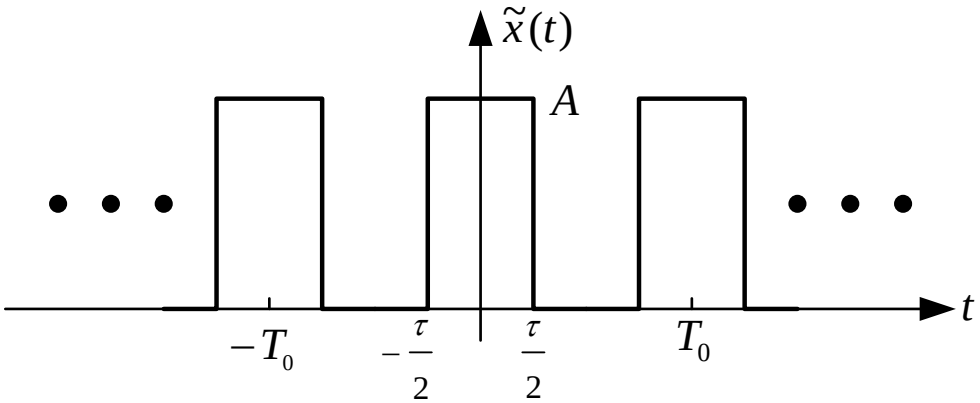
$$C_n = \frac{A\tau}{T_0}\mathrm{Sa}(\frac{n\omega_0\tau}{2})$$

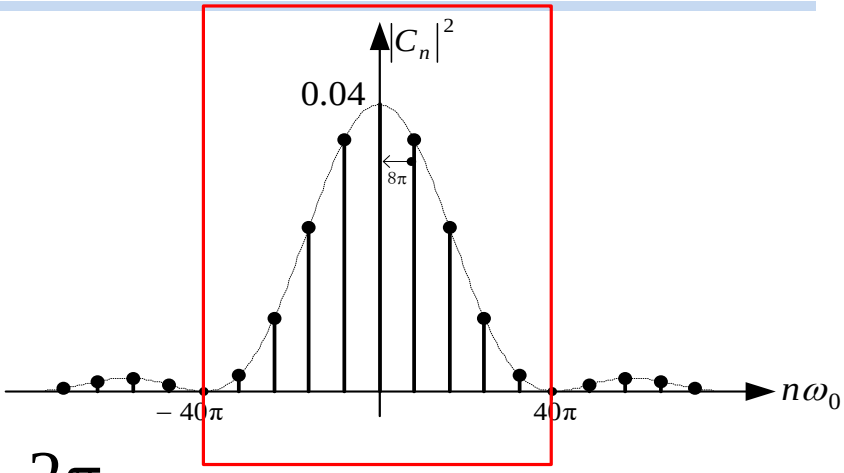
Substitute A=1，T_0=1/4，τ = 1/20，ω_0= $2\pi/T_0$ = 8π into the formula:

$$C_n = 0.2\,\mathrm{Sa}(n\omega_0/40) = 0.2\,\mathrm{Sa}(n\pi/5)$$

The power spectrum of the periodic signal: $|C_n|^2 = 0.04\mathrm{Sa}^2(n\pi/5)$

Ex.7 : Calculate the percentage of the average power of the harmonic components within the effective bandwidth ($0\sim 2\pi/\tau$) of a periodic rectangular pulse signal relative to the average power of the entire signal. Given A=1，T_0=1/4，τ=1/20.





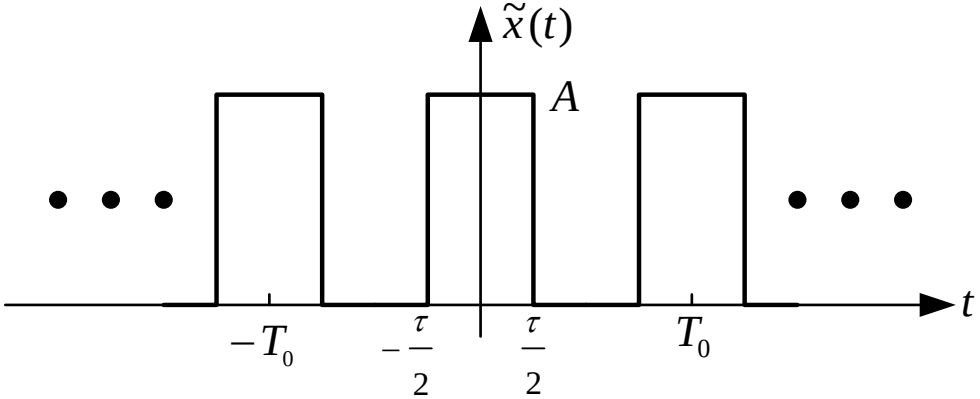
Solution: The first zero point appears at $\dfrac{2\pi}{\tau} = 40\pi = 5\times 8\pi \quad = 5\omega_0$

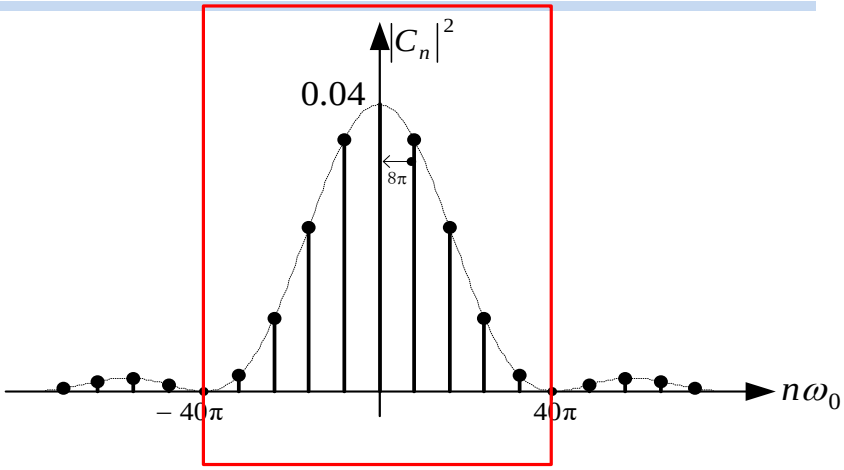
Therefore, the effective bandwidth includes the DC component and 4 harmonic components.

The power of the signal within the effective bandwidth is

$$P_1 = \sum_{n=-4}^{4} |C_n|^2 = C_0^2 + 2\sum_{n=1}^{4} |C_n|^2 = 0.1806$$

Ex.7 : Calculate the percentage of the average power of the harmonic components within the effective bandwidth ($0\sim 2\pi/\tau$) of a periodic rectangular pulse signal relative to the average power of the entire signal. Given A=1，T_0=1/4，τ=1/20.





Solution:

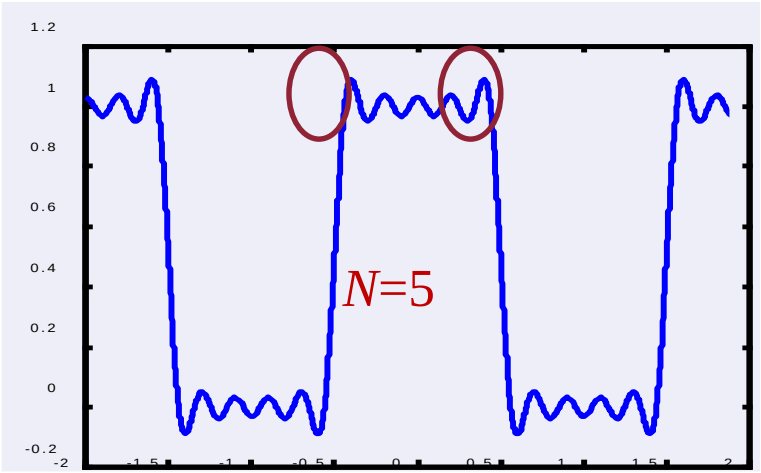
The power of the signal is $P=\dfrac{1}{T_0}\displaystyle\int_{-T_0/2}^{T_0/2}\left|\tilde{x}(t)\right|^2\mathrm{d}t=4\int_{-1/40}^{1/40}1^2\mathrm{d}t=\dfrac{1}{5}$=0.2

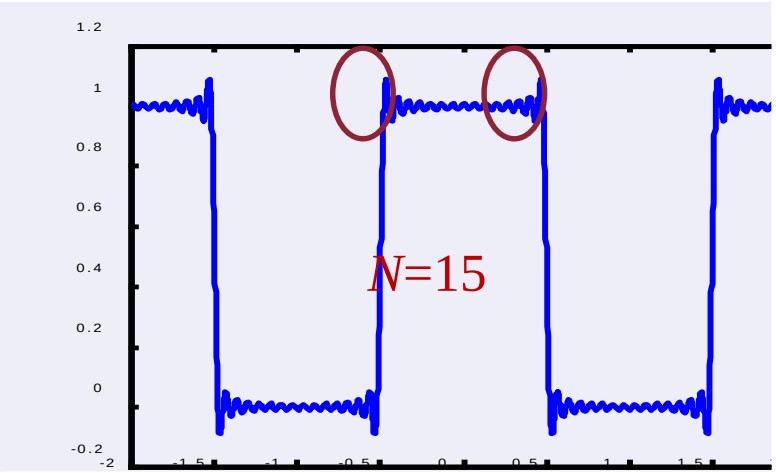
$$\frac{P_1}{P}=\frac{0.1806}{0.200}=90\%$$

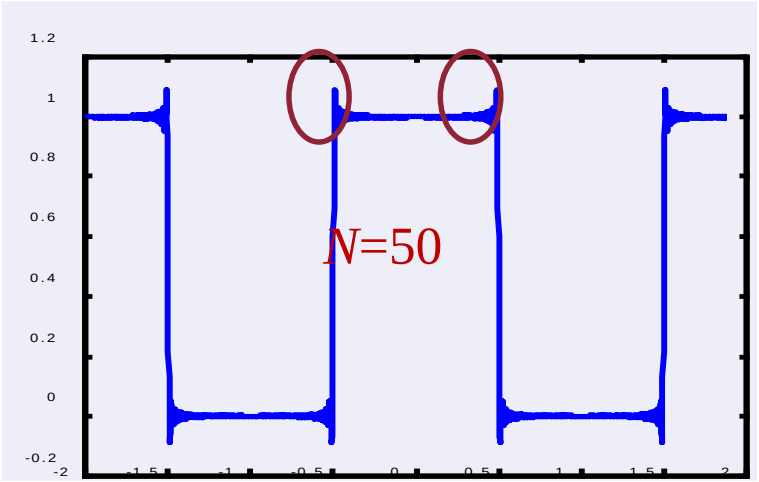
The sum of the power of all harmonic components within the effective bandwidth of the periodic rectangular signal accounts for 90% of the total signal power.

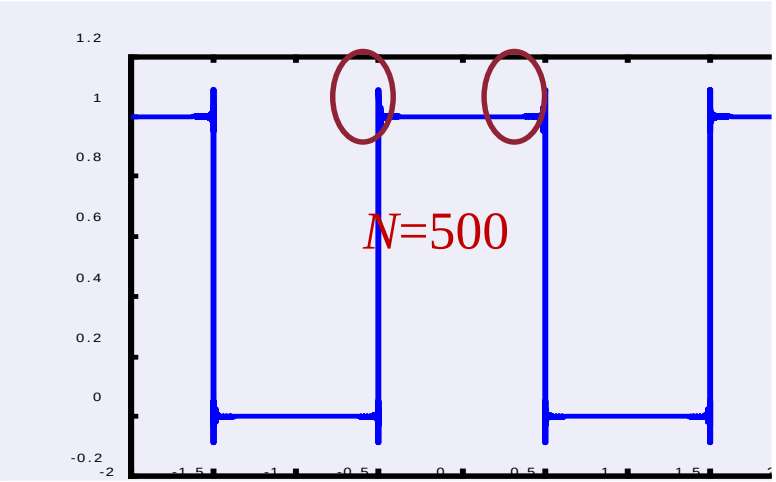
Gibbs phenomenon

$$x_N(t) = \sum_{n=-N}^{N} C_n \, \mathrm{e}^{\mathrm{j}n\omega_0 t} = C_{-N}\mathrm{e}^{-\mathrm{j}N\omega_0 t} + \ldots + C_{-1}\mathrm{e}^{-\mathrm{j}\omega_0 t} + C_0 + C_1\mathrm{e}^{\mathrm{j}\omega_0 t} + \ldots + C_N\mathrm{e}^{\mathrm{j}N\omega_0 t}$$









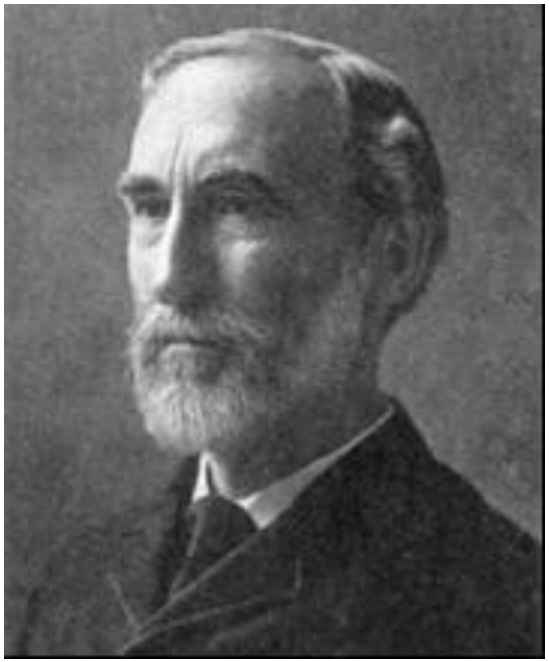
Gibbs phenomenon

Using a finite number of harmonic components to approximate the original signal, overshoot occurs at points of discontinuity. The peak value of the overshoot does not decrease with the increase of harmonic components and is 9% of the jump value.

Reason for the Gibbs Phenomenon

The existence of discontinuities in the time-domain signal disrupts the convergence of the signal, resulting in non-uniform convergence of the Fourier series at points of discontinuity.

Josiah Willard Gibbs (1839-1903)



American mathematician and physicist, professor of mathematics and physics at Yale University. In 1879, he was elected to the National Academy of Sciences and published "The Fundamental Theory of Statistics."

In 1898, American physicist Edwin Halley Mitchellson constructed a harmonic analyzer to calculate the finite-term (up to 80 terms) Fourier series approximation of any periodic signal $x(t)$. When he tested a square wave signal, he obtained a surprising result!

Near the points of discontinuity, the partial sum $x_N(t)$ exhibited oscillations, and the peak values of these oscillations did not seem to decrease with the increase of N! He began to suspect that there might be something imperfect with his equipment, so he wrote a letter to Josiah Willard Gibbs about this issue. Gibbs examined the results and confirmed their accuracy, publishing his views in 1899, which is what we now refer to as the **Gibbs phenomenon**.

Summary of Frequency Spectrum Analysis of Continuous Periodic Signals

- The mathematical tool used for analyzing problems is the Fourier series.
- The most important concept: spectrum function.
- Key points:

1. Definition and physical meaning of the spectrum.
2. Characteristics of the spectrum.
3. Properties of the spectrum, using these properties to analyze the spectrum of complex signals.
4. The concept of power spectrum and its application in engineering.

Thanks

Some of the materials cited in this course are the teaching accumulation of the main lecturer for many years, which come from a variety of media and the exchanges of colleagues, peers and friends, and it is difficult to indicate the source one by one, so I hereby explain and thank you!